

Optimizing Pit Stop Strategies with Competition in a Zero-Sum Feedback Stackelberg Game in Formula 1

Felipe Aguad

Department of Industrial Engineering, University of Chile, felipeaguad@gmail.com

Charles Thraves

Department of Industrial Engineering, University of Chile, and Institute for Complex Systems in Engineering (Sistemas Complejos de Ingeniería, ISCI), cthraves@dii.uchile.cl

Optimization of pit stop strategies in motorsports is not trivial. Most existing studies neglect competition, or account for it using simulation or historical data, but not in a game theory sense. In this work, we present a model, based on Formula 1, in which two drivers optimize their pit stop strategies. Each car decides at each lap whether to continue on-track, or to take a pit stop to change tires to one of the three tire compounds available. Since the drivers' decisions affect each other due to interactions such as overtaking, the problem is formulated as a zero-sum feedback Stackelberg game using Dynamic Programming, in which in each lap the race leader (follower) decides first (second). In addition, drivers decide simultaneously their starting tire compounds. The formulation allows for the inclusion of uncertain events, such as yellow flags, or randomness in lap times.

We define and show the existence of the game equilibrium, and provide an algorithm to find it. Then, we solve numerical instances of the problem with hundreds of millions of states. We observe how drivers' different objective functions induce different race strategies. In particular, when players maximize the probability of winning, instead of the time-gap with their opponent, their actions tend to be more risk taking. Our instances show that a strategic driver that faces another that ignores competition, increases the winning odds by more than 15% compared to when both race strategically. Finally, yellow flags tend to increase the winning chances of the driver with the worst performance.

Key words: Feedback Stackelberg, Zero-Sum Game, Dynamic Programming, Formula 1, Pit Stop Strategy

1. Introduction

Operations Research (OR) and analytical tools have widened their scope to several applications in the last decades. An example of this is its increasing presence in multiple sports, such as football, basketball, soccer, and others. More recently, racing has also started to capture the attention of researchers in parallel with practitioners, such as race engineers, who input significant amounts of effort in optimizing their race strategies (see [Fortune \(2015\)](#)). Although optimization of race strategies involves a myriad of variables, such as the choice of laps in which to do a pit stop, tire compounds, fuel management, car set-up, and more, we focus on the first two elements of the list in our work. In particular, we frame

the problem in a Formula 1 context; nonetheless, the methodology can also be applied to other racing categories.

The result of a race depends on several factors besides the car and driver performance. Some of these factors are uncertain, such as car crashes or engine failures, whereas others are controlled by the racing engineers, as for example the laps in which to have a pit stop and the initial tire compound at the start of the race. In addition, uncertain events during the race, such as yellow flags, make pit stop strategy even more important. The two main yellow flag events in Formula 1 are: (i) Virtual Safety Car (VSC), in which cars must decrease their speed while maintaining the time difference to their rivals, and (ii) Safety Car (SC), in which case cars are queued behind the safety car at a slow speed. Yellow flags are usually triggered by racing incidents that must be cleared. An example of this, is how Red Bull Racing pitted Max Verstappen in the Abu Dhabi Grand Prix (GP) in 2021 during the yellow flag that occurred 5 laps before the end, allowing him to replace a used set of hard compound tires with a new set of soft ones. This allowed him to overtake Lewis Hamilton (who did not enter the pit to avoid losing his track position), winning the race and championship in the final lap. (See ([Formula 1 2021](#)) for more details). It is worth mentioning that refueling has no longer been a practice in Formula 1 since 2010.

Tires degrade as they get used, resulting in a decline in performance in terms of lap times. To counter this, a pit stop provides an opportunity to replace the current set of worn tires with new ones. Consequently, lap times improve after a pit stop; however, the process itself takes approximately 30 seconds, with a loss of about 20 seconds relative to the other competitors who did not stop. This introduces a trade-off between the number of pit stops, and the level of tire degradation. Another factor to consider is the selection of tire compounds for each *stint*. A stint refers to a consecutive set of laps during which the same set of tires is utilized. For example, if a car pits in laps 10 and 25 in a race with a total of 30 laps, we would say that the car has three stints of 10, 15, and 5 laps, respectively. In Formula 1, in 2023, there are six tire compounds for dry weather, but only three are permitted for use by all drivers during a race. These compounds are categorized as *soft*, *medium*, and *hard*. While softer compounds offer faster lap times, they also degrade more quickly compared to the harder compounds. As a result, the car's performance deteriorates rapidly when using soft compound tires. On the other hand, hard compounds have a longer lifespan but provide slightly inferior performance, particularly when the tires are new.

Hence, a car should choose between using softer tires in shorter stints, or using harder tires with less stops. For a visual explanation of this trade-off, see [Chain Bear \(2017\)](#).

The cost in terms of time loss during a pit stop is not independent of the state of the race. Random events, such as a yellow flag, enforce all cars to slow down. Consequently, opting for a pit stop under a yellow flag presents a time advantage compared to doing so under normal race conditions. As a result, strategic behaviors are occasionally observed, such as extending the tire usage in races where the likelihood of a pit stop is high.

Another crucial factor to consider is the presence of interaction and competition among drivers. In terms of interaction, there are various ways in which cars influence each other, thereby impacting their respective lap times. For instance, when one driver overtakes another, their lap times are affected. Similarly, a driver defending its position against a potential overtake, or closely trailing another car in front¹ can affect lap times. Consequently, the lap time of a car is influenced not only by internal factors, such as the tire compound or tire usage, but also by external factors that involve interactions between cars. Hence, drivers' actions, particularly their pit stop strategies, should take these interactions into account. Additionally, in terms of competition, each driver should optimize its pit stop strategy taking the actions of other drivers into consideration as well, in a game theory sense.

In this work we present a game theory model in which two drivers optimize their pit stop strategies during the race. More specifically, during each lap, both drivers decide whether to continue on track, or change their tires to a specific tire compound. Every lap is modeled as a Stackelberg game in which the first driver plays first. Thus, the leader in each stage is not fixed, as they could alternate depending on the driver who is leading the race. In addition to the sequential structure of the game during the race, at the start, the drivers choose the tire compound to start the race with in a simultaneous game. Unfortunately, even in a deterministic setting, i.e., no yellow flags events, and no uncertainty in lap times, the extended form of the game has an exponential number of nodes. Indeed, consider a race with 50 laps and three tire compounds. At each lap drivers have four possible actions: change tires to one of the three available tire compounds, or continue on track. In this

¹ This could affect the follower's lap times. More specifically, following a leading car in turns increases the lap time of the former because the car in front removes the air resistance (i.e., downforce) of the follower. Contrarily, following in straights has the opposite effect on the follower.

example, the total number of terminal nodes is equal to $4^{50} \cdot 4^{50}$, which makes it impractical to solve in a reasonable time. However, each sequence of actions of both drivers until each lap can be reduced to a *state*. As a result, we propose to model the game as a feedback Stackelberg game, and solve it with dynamic programming. The main contributions of this work can be summarized in the following points:

- We formulate the pit stop strategy optimization problem with competition for two drivers using a feedback Stackelberg zero-sum game. The model allows incorporating stochastic scenarios which include yellow flag events, such as virtual safety car or safety car, and uncertainty in the cars' lap times and interactions. In addition, the game incorporates the initial tire decisions which are played simultaneously. The model also allows different types of objective functions, such as the maximization of the time-gap with respect to the other driver, or maximization of the odds of winning.
- We define the feedback Stackelberg equilibrium of the game which involves the players' pit stop strategies, as well as the Nash-feedback Stackelberg equilibrium which adds the initial tire strategy. We show the existence of these equilibria, and formulate a backward induction algorithm with dynamic programming that is played sequentially by both drivers for all the laps of the race. We prove that our proposed algorithm finds the game equilibrium.
- We implement a code able to solve in the order of hours instances of the problem with hundreds of millions of states in each lap. We show that strategic decision making enhances players' race outcome. Indeed, we observe instances in which a driver's winning odds increase by more than 15% when the opponent is not strategic. In addition, we show how players' decisions change when using different objective functions.

In Section 2 we review the related literature to this work. Section 3 shows a deterministic model, and Section 4 extends it to a stochastic setting. Results of numerical computations are provided in Section 5. Finally, conclusions are stated in Section 6. Proofs are relegated to the Appendix.

2. Related Literature

Optimization of race strategies has been explored from different perspectives. McLaren Racing Limited (2019) analyzed the optimization of fuel during the race, which includes the initial amount of fuel at the start of the race, and the lap and amount of fuel to refuel

in a pit-stop. Optimization of battery has also been a subject of study by (Liu et al. 2021, Anselma 2022), where decision variables are the depletion and thermal management of the battery.

Since the last 15 years, several works have focused on the pit stop strategy optimization problem. Bekker and Lotz (2009) used simulation to model passing maneuvers, pit stops, and car failures; generating race outcomes given an input set of parameters that describe the race. By evaluating different pit stop strategies, which are inputs of the simulation model, the authors can find the optimal race strategy. In a similar direction Phillips (2014) developed a simulator that considers several inputs of the drivers, including their performance in one of the free practices², tire degradation, lap-time variability, and others. Also using simulation, Heilmeier et al. (2018) extended the previous work by accounting for additional elements, such as using times of other practice sessions and the qualifying session³, and checking the time gap between cars in multiple parts of the track (rather than just at the end of the lap). Heilmeier et al. (2020a) generalizes the race simulation by considering uncertainty in different aspects, such as lap times, occurrence of yellow flag events, pit stop duration, accidents, and failures. F1 Strategy Blog (2019) created an open-source code that simulates the race. In this, race strategies of previous years are used. Although simulation techniques are useful for incorporating several variables of the problem, their main problem is the difficulty of finding optimal pit stop strategies that consider the competitive race dynamics.

Machine learning techniques have also been used to predict car performance, including the pit stop strategy as one of the explanatory variables used. Under this framework, Tulabandhula and Rudin (2014) and Choo (2015) used machine learning techniques trained on previous races to predict lap times using variables such as making a pit stop, car performance, and others. More recently, Heilmeier et al. (2020b) combined artificial neural networks (NN) and simulation to find the best race strategy for each driver. The NN are trained on race data of previous seasons, predicting whether or not to make a pit stop and which tire allocation to use (if pitting). Then, the race is simulated so that at each lap drivers' pit stop decisions are given by the predictions of the NN. One of the inputs

² Each race has three practice sessions before the race where drivers are allowed to test the circuit. The practice data used as input in Phillips (2014) is the seconds practice.

³ The positions of the starting grid of the race are determined during the qualifying session. See The Sport Rush (2020) for more details.

used in their NN is the number of remaining pit stops to make in the race, which is found by solving a mixed-integer quadratic program assuming there is no interaction between cars nor uncertain events such as yellow flags. Our work differs from this since we propose a method to compute the optimal strategies of two competing drivers in a game theory setting, that is, the decisions are based on the optimal strategies and not on prediction methods calibrated on the decisions of past races.

One of the most similar works to this one is the work in [Heine and Thraves \(2022\)](#). The authors present a dynamic programming solution approach to find the optimal pit stop strategy in a model that accounts for VSC events in a setting with no competition or interaction between drivers. The main difference of [Heine and Thraves \(2022\)](#) and this work is the inclusion of competition between two drivers. In particular, the incorporation of interaction events between cars, while also accounting for the game theory aspect in the decision making of both players. In particular, we use a zero-sum *feedback* Stackelberg game with finite action sets and time horizon. Feedback Stackelberg games are characterized by having multiple stages which in each stage all the players know the state at which the game has evolved, and also, there is a player who acts as a leader in each of the stages. This type of game was first introduced in [Simaan and Cruz \(1973a\)](#) and [Simaan and Cruz \(1973b\)](#). Applications of Feedback Stackelberg games in Operations Management are shown in [Li and Sethi \(2017\)](#). For more details see [Başar and Olsder \(1998\)](#).

An interesting aspect of the problem addressed here is the alternation of the leader role at different stages (laps) of the game, depending on who is leading the race at the current lap (unlike most feedback Stackelberg settings in which the same player acts as leader in every stage of the game). One of the few studies that have addressed situations with alternating leaders is [Nie et al. \(2008\)](#). In this work, the authors presented a dynamic programming framework to solve dynamic feedback Stackelberg games in discrete time, in which the leader at each stage depends on the players' payoff function at that specific stage. In our case, the leader of each stage will depend on a component of the state variable. Also, our game setting accounts for stochastic events.

3. Deterministic Model

In this section we present a sequential game modelled with Dynamic Programming in which players (drivers) make decisions at each stage (lap) depending on the current state (tire

compound allocation, tire usage, time gap between drivers, etc). Then, we provide details on the computation of lap times. Summary tables for the notation used are provided in Appendix A.

Consider two drivers, A and B , that compete in a race of N laps. In each lap $n \in \{1, \dots, N\}$, the drivers decide whether to continue on track or to stop to change tires to one of the tire compounds in the set $\mathcal{T} = \{1, 2, 3\}$; where 1=Soft, 2=Medium, and 3=Hard. We assume tire $t \in \mathcal{T}$ has a lifespan of u_t laps. Thus, a tire compound t can not be used for more than u_t laps. We denote by $x_n^A, x_n^B \in \mathcal{T}_0 := \{0\} \cup \mathcal{T}$ the decisions of drivers A and B , respectively, in lap n . Our convention is that these variables take value zero if there is no pit stop. Naturally, the sequence of decisions within a lap is such that the leading car decides first. Note that in this sequential game the order in which drivers play at each stage, i.e, lap, might not be the same during the whole race. For example, if driver A starts the race first, but at some point is overtaken by his opponent, A will go from being a leader (in a game theory sense) to a follower.

Without loss of generality, it is assumed that the pit stop entrance is just after the start/finish line of the race⁴, see Figure 1. Then, once the driver who is leading the race observes the state variables (which will be defined below) at the beginning of a lap, he decides the action to take. Afterwards, the driver who is second decides an action assuming known the state variables as well as the decision of his opponent who has already played. Here we are assuming that the car that is second observes not just if the driver in front stops (which is reasonable to assume), but also what will be his new tire compound. We believe this is reasonable since the driver in second place could compute the best strategy of the leading driver given the current state.

⁴ In practice, the pit stop entrance is a few meters before the finish line, and its exit is a few hundred meters after this.

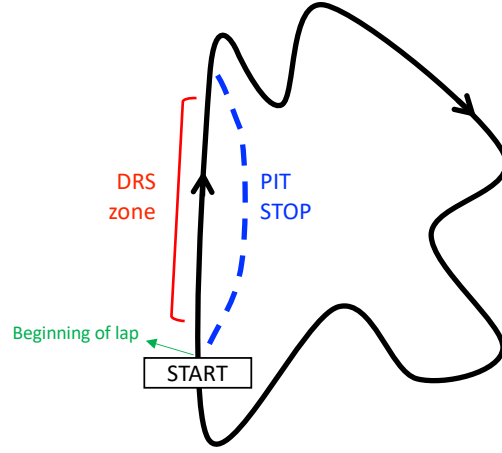


Figure 1 Circuit.

The state at the beginning of lap n is described by variables that depend on the state of each car, as well as on elements that depend on both cars. For each driver $i \in \{A, B\}$ in the beginning of lap n , we have the following state components: the current tire compound denoted by t_n^i , the usage of the tires denoted by w_n^i in units of laps, and a binary indicator variable m_n^i which takes value 0 if the driver has used only one tire compound, and 1 otherwise. This indicator variable allows us to know if the car that ends the race satisfies the Formula 1 rule that states that at least two different tire compounds should be used during the race. (See [MotorSports \(2022\)](#) for more details). The last state variable component is the time difference between drivers A and B , which is denoted by g_n . Our convention is that a negative value indicates that driver A is ahead of driver B by $|g_n|$ seconds. With respect to the tire wear state variable, w_n , its maximum value is set to $u_{t_n} + 1$, i.e., one additional lap to the tire lifespan, to indicate the case where the tire life exceeds its maximum usage, therefore, the car cannot continue. If this happens, we assume that all future actions of the driver are set to 0. Putting together all the state variable components, the state at lap n is denoted by:

$$s_n = (t_n^A, w_n^A, m_n^A, t_n^B, w_n^B, m_n^B, g_n). \quad (1)$$

The transition of the state variables from one stage to the next as a function of the previous state variables and drivers' actions, i.e., $s_{n+1}(s_n, x_n^A, x_n^B)$, can be written in the following expressions:

$$t_{n+1}^A = t_n^A \cdot \mathbb{1}_{\{x_n^A=0\}} + x_n^A \cdot \mathbb{1}_{\{x_n^A \neq 0\}} \quad (2)$$

$$w_{n+1}^A = \min\{w_n^A \cdot \mathbb{1}_{\{x_n^A=0\}} + 1, u_{t_n} + 1\} \quad (3)$$

$$m_{n+1}^A = \max\{m_n^A, \mathbb{1}_{\{x_n^A \neq 0, x_n^A \neq t_n^A\}}\} \quad (4)$$

$$g_{n+1} = g_n + \mu_n^A(s_n, x_n^A, x_n^B) - \mu_n^B(s_n, x_n^A, x_n^B), \quad (5)$$

where μ_n^A and μ_n^B are the lap time functions of drivers A and B , respectively. (More details on these are provided in the next section). We omit the transition equations for t_{n+1}^B , w_{n+1}^B , and m_{n+1}^B since these are analogous to those of Equations (2), (3), and (4). Equation (2) states that the tire compound of the next lap will be the same as in the previous lap in the case of not pitting, otherwise this will be the compound installed on the car during the pit stop. Equation (3) states that the tire usage increases by one unit from one lap to the next one, except when a pit stop occurs. In case of a pit stop, the tire usage at the beginning of the following lap will be equal to one. In Equation (4), the indicator variable of the next lap, m_{n+1} , is the minimum between its previous value, and 1 in case of changing tires to a different compound. Then, this state variable is non-decreasing and can only take values in $\{0, 1\}$. The update on the time difference between cars is given by Equation (5). In particular, the time gap in the next lap is equal to the previous time gap plus the lap time difference between the two cars during the current laps.

The goal of each player is to end the race with the maximum time difference to their opponent. Then, the game can be modelled as a zero-sum game where the *utility* function is the time difference between the two cars when crossing the finish line at lap N . Namely, g_{N+1} (the index $N + 1$ in this case means “at the start of the hypothetical $N + 1$ lap, or equivalently, at the very end of lap N ”). Our convention is that this time difference reflects the number of seconds A is ahead of B . Driver A will minimize the utility function, while B is going to maximize it. The utility function at the end of the race should account for cases when one or both of the variables m_{N+1}^A and m_{N+1}^B are zero. In this case, the corresponding car is disqualified. Then, the border condition can be summarized as follows:

$$V_{N+1}^*(s_{N+1}) = \begin{cases} g_{N+1} & \text{if } m_{N+1}^A = 1 \text{ and } m_{N+1}^B = 1 \\ -\infty & \text{if } m_{N+1}^A = 1 \text{ and } m_{N+1}^B = 0 \\ +\infty & \text{if } m_{N+1}^A = 0 \text{ and } m_{N+1}^B = 1 \\ 0 & \text{otherwise.} \end{cases} \quad (6)$$

Then, the Bellman Equation can be expressed as:

$$V_n(s_n, x_n^A, x_n^B) = V_{n+1}^*(s_{n+1}), \quad (7)$$

where for ease of exposition we have omitted the explicit dependency of the future state components (s_{n+1}) on the previous state and decisions $(s_n, x_n^A$ and $x_n^B)$. For $n = 2, \dots, N$

$$V_n^*(s_n) = \begin{cases} \min_{a \in \mathcal{T}_0} V_n(s_n, a, x_n^B(s_n, a)) & \text{if } g_n < 0 \\ \max_{b \in \mathcal{T}_0} V_n(s_n, x_n^A(s_n, b), b) & \text{if } g_n > 0, \end{cases} \quad (8)$$

where

$$x_n^A(s_n, b) = \arg \min_{a \in \mathcal{T}_0} V_n(s_n, a, b), \quad (9)$$

$$x_n^B(s_n, a) = \arg \max_{b \in \mathcal{T}_0} V_n(s_n, a, b), \quad (10)$$

with abuse of notation as we use $x_n^A(s_n, \cdot)$ and $x_n^B(s_n, \cdot)$ to denote the best response function of the driver who acts as the follower at lap n (A if $g_n > 0$, B otherwise) given the state variable s_n . We impose there is no pit stop in the first lap (due to the modelling of the entrance of the pit stop), and in the last lap (since a car cannot end the race entering the pit lane); i.e., $x_1^A = x_1^B = x_N^A = x_N^B = 0$. The state variable at the beginning of the race is:

$$\begin{aligned} s_1 &= (t_1^A, w_1^A, m_1^A, t_1^B, w_1^B, m_1^B, g_1) \\ &= (t_1^A, 0, 0, t_1^B, 0, 0, g_1), \end{aligned} \quad (11)$$

where t_1^A and t_1^B are the tire compounds at the start of the race, and g_1 is the number of seconds A is ahead of B when A crosses the finish⁵ line at the start of the race. We assume that this quantity, g_1 , is known since it can be estimated considering the starting position on the grid of both drivers. As for the initial tire compounds, these are decided by both drivers before the race. Unlike the sequential nature of pit stop decisions during the race, the initial tire compounds can be modelled as a simultaneous game.

Consequently, once we have the equilibrium of players' pit stop strategies for each initial tire compound combination (i.e., all elements in $\mathcal{T} \times \mathcal{T}$), we can solve the simultaneous game to find the equilibrium of the initial tire compounds. (The formal definition of the game Equilibrium and the proof of its existence is going to be shown in the next section). More precisely, we can find the Equilibrium in mixed strategies of the zero-sum game in which the players' action space is the finite set $\mathcal{T}$ where the pay-off matrix is given by $U \in \mathbb{R}^{|\mathcal{T}| \times |\mathcal{T}|}$ where $U_{t_1^A, t_1^B} := V_1^A((t_1^A, 0, 0, t_1^B, 0, 0, g_1))$ for $t_1^A, t_1^B \in \mathcal{T}$. At this stage, at the

⁵ The starting position of the cars is behind the finish line. The *starting grid* is the order in which the drivers start the race.

beginning of the race, the equilibrium can be solved by using the Lemke-Howson algorithm, see [Von Stengel \(2002\)](#). However, since the size of the problem at this stage is small (as both players have only three strategies), we can use Linear Programming (LP).

Let us denote by $\pi^A, \pi^B \in \Delta^{|\mathcal{T}|-1}$ the mixed strategies of the two players, respectively, where Δ^n is the $n + 1$ dimensional simplex. The LP formulations to find the mixed equilibrium is provided in [Appendix B](#).

3.1. Lap Time Function

The only missing element in the Dynamic Programming is the definition of the lap time functions, μ_n^A and μ_n^B , that appear on the right-hand side of Equation (5). In the following we will present the lap time function of only driver A , since the function for driver B is analogous. The lap time is expressed as a function of three elements: (i) a *ghost lap time function*, which represents the lap time of a car given the hypothetical situation that he is driving alone (i.e., no competition); (ii) an interaction function, that includes the time gained/lost by being close either from behind or in front of the opponent's car; and (iii) time reduction due to the use of the Drag Reduction System (DRS)⁶. Drivers can activate the DRS when the car in front of them has a time gap of less than a second. We assume that this DRS zone is in the straight of the track in parallel to the pits, see [Figure 1](#). Then, the lap time function for driver A can be written as:

$$\mu_n^A(s_n, x_n^A, x_n^B) = \gamma_n^A + \eta^A - t^{DRS} \cdot \mathbb{1}_{\{g_n \in [0,1]\}}, \quad (12)$$

where each term of the RHS of Equation (12) corresponds to (i), (ii), and (iii) of this paragraph, respectively. In particular, γ_n^A denotes the *ghost lap time function*, η^A is the *interaction function* which denotes the additional lap time caused by the interaction between the drivers, and t^{DRS} denotes the parameter that represents the time reduction due to the use of the DRS. For ease of exposition we omit the explicit dependency of these functions on the state and decision variables.

The *ghost lap time function* γ_n^A can be written as:

$$\gamma_n^A = d_0^A + p_0^A \cdot \mathbb{1}_{\{x_n^A \neq 0\}} + \beta(t_{n+1}^A(t_n^A, x_n^A), w_n^A) - h \cdot n, \quad (13)$$

⁶ The DRS main task is to reduce air resistance by opening up the rear wing and therefore reducing the downforce. This provides an extra 10-15 km/h speed on straights. The places in which the DRS can be used are called *DRS zones*.

where d_0^A is the base lap time of A assuming he is driving with a new set of soft compound tires during the first lap of the race, and assuming no interaction with other cars. p_0^A represents the relative additional time spent for a pit stop compared to a scenario in which a pit stop is not made. The function β captures the relation between the tire compound, its usage, and the corresponding increase in lap time compared to using a new soft compound tire. Note that we have used Equation (2) to get the actual tire compound with which the driver will run the n^{th} lap, since the tires used for the n^{th} lap are going to depend on the current tires at the beginning of this lap (t_n^A), and on the pit stop decision (x_n^A). Finally, the parameter h represents the lap time reduction for each additional lap that is completed, all else being equal. This accounts for the fact that since the fuel is consumed during the race, the weight of the car decreases as more laps are raced. We assume this effect is linear, nonetheless, we could also use more elaborated functional forms to model this effect.

The *interaction function* η^A models the additional lap time for driver A caused by the interaction between cars from the pit stop exit to the end of the lap. Note that although there could be a wide time gap between cars at the beginning of a lap n (that is, $|g_n|$ is large), this could be decreased if the leading car stops while the following car continues on track. In such a case, the cars could end up very close to each other after the pit stop exit despite being far apart at the start of the lap. We use ϵ_n to denote the time difference between the cars at lap n , similar to the state variable g_n , but at the pit exit. This can be written as:

$$\epsilon_n = g_n + p_0^A \cdot \mathbb{1}_{\{x_n^A \neq 0\}} - t^{DRS} \cdot \mathbb{1}_{\{g_n \in [0,1]\}} - (p_0^B \cdot \mathbb{1}_{\{x_n^B \neq 0\}} - t^{DRS} \cdot \mathbb{1}_{\{g_n \in [-1,0]\}}). \quad (14)$$

Then, the *interaction function* is written as a function of ϵ_n . More precisely,

$$\eta^A(\epsilon_n) = \begin{cases} \exp(-\lambda \cdot |\epsilon_n|) \cdot z^{(1)} & \text{if } \epsilon_n < 0 \\ \exp(-\lambda \cdot |\epsilon_n|) \cdot z^{(2)} & \text{otherwise,} \end{cases} \quad (15)$$

where $\lambda, z^{(1)}, z^{(2)} > 0$, are parameters of the *interaction function* for car A . The idea behind using different parameters, $z^{(1)}$ and $z^{(2)}$, depending on the opponent's position relative to the car (i.e., whether it's in front or behind it) is to account for the potential variation in time loss. Being closely followed by a car may have a different impact than having the same time difference but led by the opponent. Additionally, the exponential function is

used to ensure that the interaction effect diminishes as the gap between the cars increases.

The *interaction function* for car B , η^B is the same as in Equation (15), except that the first condition should be changed to $\epsilon_n > 0$.

3.2. Algorithm

The summary of the steps to solve the game are provided in Algorithm 1. In particular, we solve the feedback Stackelberg game first by solving the DP for all laps of the race in reverse order for both players, and then, solve the simultaneous game before the start of the race to obtain the equilibrium in mixed strategies regarding the tire compounds with which to start the race. We denote by $\mathcal{S}_n$ the set of all states at lap n .

Algorithm 1 Dynamic Programming Algorithm

```

1: Input: Set of states  $\{\mathcal{S}_n\}_{n \in \{1, \dots, N+1\}}$ . Starting time gap  $g_1$ 
2: Output: Decisions  $\{x_n^{A*}(s_n)\}_{s_n \in \mathcal{S}_n, n \in \{1, \dots, N\}}$ ,  $\{x_n^{B*}(s_n)\}_{s_n \in \mathcal{S}_n, n \in \{1, \dots, N\}}$ ,  $\pi^{A*}$  and  $\pi^{B*}$ .
3: Foreach  $s_n \in \mathcal{S}_{N+1}$ 
4:   Compute  $V_{N+1}^*(s_n)$  using Equation (6).
5: End For
6: Set  $n \leftarrow N$ 
7: While  $n \geq 1$  do
8:   Foreach  $s_n \in \mathcal{S}_n$ 
9:     Compute  $V_n(s_n, a, b)$  from Equation (7) for all  $a, b \in \mathcal{T}_0$ .
10:    If  $g_n < 0$  Then
11:      Compute  $x_n^B(s_n, a)$  with Equation (10) for all  $a \in \mathcal{T}_0$ .
12:       $x_n^A(s_n) \leftarrow \arg \max_{a \in \mathcal{T}_0} V_n(s_n, a, x_n^B(s_n, a))$ .
13:       $V_n^*(s_n) \leftarrow V_n(s_n, x_n^A(s_n), x_n^B(s_n, x_n^A(s_n)))$ .
14:    Else
15:      Compute  $x_n^A(s_n, b)$  with Equation (9) for all  $b \in \mathcal{T}_0$ .
16:       $x_n^B(s_n) \leftarrow \arg \max_{b \in \mathcal{T}_0} V_n(s_n, x_n^A(s_n, b), b)$ .
17:       $V_n^*(s_n) \leftarrow V_n(s_n, x_n^A(s_n, x_n^B(s_n)), x_n^B(s_n))$ .
18:    End If
19:  End For
20:   $n \leftarrow n - 1$ 
21: End While
22: Compute  $U \in \mathbb{R}^{|\mathcal{T}| \times |\mathcal{T}|}$  such that  $U_{t_1^A, t_1^B} = V_1^A((t_1^A, 0, 0, t_1^B, 0, 0, g_1))$  for  $t_1^A, t_1^B \in \mathcal{T}$ .
23: Get  $\pi^{A*}$  and  $\pi^{B*}$  by solving Optimization Problems (46) and (47) in Appendix B.

```

4. Stochastic Model

The model introduced in the previous section is deterministic; as a result, the race outcome is determined uniquely by the players' actions. In reality, there are several uncertain events that might unfold during the race, such as yellow flags or variation in lap times. In this section, a dynamic programming formulation for the game that accounts for these elements is presented. The section is divided into three parts. Section 4.1 describes the modelling of yellow flags. Section 4.2 shows the details on the computation of lap times.

Finally, Section 4.3 provides the formulation of the stochastic dynamic programming, the equilibrium definition, its existence, and an algorithm that finds it. The description of the random variables are introduced in Sections 4.1 and 4.2.

4.1. Yellow Flags

The two yellow flag events considered are Virtual Safety Car (VSC), and Safety Car (SC). (We use the expression “safety car”, without acronym, to refer explicitly to the car that is deployed on the track to enforce the speed reduction of the drivers). In both cases, there is a racing incident that involves clearing the track either of debris, or a damaged car. In VSC, drivers are forced to reduce their speed by 40% (see Biswas (2022)), whereas in SC, drivers should follow the safety car deployed on the track. In this case cars are lined up behind the safety car, and therefore time gaps are reduced. In either VSC or SC, overtakes on track are not allowed, but overtakes produced as a result of pit stops are permitted. Let Y_n^{VSC} and Y_n^{SC} be the random variables at lap n that denote the remaining laps (without considering lap n) under VSC and SC, respectively. For simplicity, we assume that the yellow flag incident occurs at the end of the lap. We assume a VSC (SC) incident lasts for l^{VSC} (l^{SC}) laps. Before defining the probability law of the random variables, let us define the state variables.

In order to know if the current lap is being raced under a yellow flag event, two additional state variable components are considered (in addition to the ones introduced in the DP of Section 3) that account for this fact. In particular, for each lap n , let y_n^{VSC} (y_n^{SC}) denote the remaining laps under VSC (SC). By the definition of the random variables (Y_n^{VSC}, Y_n^{SC}) and state variable components (y_n^{VSC}, y_n^{SC}), it is clear that $y_{n+1}^{VSC} = Y_n^{VSC}$ and $y_{n+1}^{SC} = Y_n^{SC}$. Then, the probability of a yellow flag event is defined as follows:

$$\begin{aligned} \mathbb{P}((Y_n^{VSC}, Y_n^{SC}) = (i, 0) \mid (y_n^{VSC}, y_n^{SC}) = (i+1, 0)) &= 1 \quad \forall i \in \{1, \dots, l^{VSC}\} \\ \mathbb{P}((Y_n^{VSC}, Y_n^{SC}) = (0, i) \mid (y_n^{VSC}, y_n^{SC}) = (0, i+1)) &= 1 \quad \forall i \in \{1, \dots, l^{SC}\} \\ \mathbb{P}((Y_n^{VSC}, Y_n^{SC}) = (l^{VSC}, 0) \mid (y_n^{VSC}, y_n^{SC}) = a) &= q^{VSC} \quad \forall a \in \{(0, 0), (1, 0), (0, 1)\} \\ \mathbb{P}((Y_n^{VSC}, Y_n^{SC}) = (0, l^{SC}) \mid (y_n^{VSC}, y_n^{SC}) = a) &= q^{SC} \quad \forall a \in \{(0, 0), (1, 0), (0, 1)\}, \end{aligned}$$

where $q^{VSC}, q^{SC} \in [0, 1]$, and PMF 0 for other cases. Namely, if there are j laps left under a VSC, with $j > 1$, then with probability 1 in the next lap there will be $j - 1$ laps remaining

under the same VSC event⁷. However, if there is no yellow flag (i.e., $(y_n^{VSC} + y_n^{SC} = 0)$, or a yellow flag is ending (i.e., $(y_n^{VSC} + y_n^{SC} = 1)$, then with probability q^{VSC} (q^{SC}) a VSC (SC) event happens at the end of lap n . Note that Y_n^{VSC} and Y_n^{SC} are not independent, because VSC and SC cannot occur simultaneously. For ease of notation, we denote by r^{VSC} (r^{SC}) the probability of having at least one VSC (SC) during the race. Thus, $r^v = 1 - (1 - q^v)^{1/N}$ for $v \in \{VSC, SC\}$.

In addition to the two state variable components, y_n^{VSC} and y_n^{SC} , there is a third state component that is included for the stochastic setting. As mentioned above, after a yellow flag event, in particular under a SC, time gaps between cars are significantly reduced. As a result, the DRS is disabled during the yellow flag, and is enabled a couple of laps after the yellow flag finishes. (This is to ensure that the DRS is not enabled when the cars are very close to each other.) Also, for similar reasons, the DRS is disabled during the first two laps of a race. We denote by k^{VSC} (k^{SC}) the number of laps after the end of a VSC (SC) in which the DRS is enabled. Let y_n^{DRS} be a state variable component that denotes the remaining laps in which the DRS will be enabled, taking value 0 if the DRS is enabled in the current lap. Then, the state variable at lap n , s_n , is conformed by:

$$s_n = (t_n^A, w_n^A, m_n^A, t_n^B, w_n^B, m_n^B, g_n, y_n^{VSC}, y_n^{SC}, y_n^{DRS}). \quad (16)$$

The impact of yellow flags in lap times is explained in the following section.

4.2. Lap times

Yellow flags affect the speed of the cars and therefore their lap times. We provide lap time expressions for the three following scenarios: (i) being under a VSC, (ii) being under a SC, and (iii) not being under a yellow flag. We denote the lap time functions under these three settings by $\mu_{n,VSC}^i$, $\mu_{n,SC}^i$, and $\mu_{n,0}^i$ for each driver $i \in \{A, B\}$ in lap n , respectively.

4.2.1. Lap time under VSC: On the one hand, under a VSC every driver has to reduce their speed. Then, lap times in this case can be expressed as:

$$\mu_{n,VSC}^i = d_{VSC} + p_{VSC}^i \cdot \mathbb{1}_{\{x_n^i \neq 0\}}, \quad (17)$$

with $i \in \{A, B\}$, where d_{VSC} is the lap time of a car during a VSC when there is no pit stop, and p_{VSC}^i is the additional time loss by driver i because of doing a pit stop during a VSC.

⁷ Note that by “laps remaining under VSC” (and similarly SC), we mean the number of remaining laps under a particular VSC, assuming there are no more yellow flag events in the future.

4.2.2. Lap time under SC: On the other hand, under a SC, drivers must follow at a slow pace a safety car deployed after the pit stop exit. We assume that the time difference between cars is decreased to 0.5 seconds. More formally, consider the function ξ that takes value 1 if driver A ends up ahead of B after the pit stop exit of a particular lap, and 0 otherwise. Namely,

$$\xi(g_n, x_n^A, x_n^B) = \mathbb{1} \{g_n + p_{SC}^A \cdot \mathbb{1}_{\{x_n^A \neq 0\}} - p_{SC}^B \cdot \mathbb{1}_{\{x_n^B \neq 0\}} < 0\}, \quad (18)$$

where p_{SC}^i denotes the additional time lost by driver $i \in \{A, B\}$ due to a pit stop while being under a SC. Then, under a SC, the state variable g_{n+1} can be written as follows:

$$g_{n+1} = 0.5 \cdot (-1)^{\xi(g_n, x_n^A, x_n^B)}. \quad (19)$$

Equation (19) shows that while the SC is on track, one of the drivers, either A or B , will be leading the race by 0.5 seconds. As a result of the non-overtaking on track rule during a yellow flag, the only way an overtake happens is because of a pit stop. This would take place if $g_n < 0$ and $\xi(g_n, x_n^A, x_n^B) = 0$ (thus $g_{n+1} = 0.5$), or $g_n > 0$ and $\xi(g_n, x_n^A, x_n^B) = 1$ (thus $g_{n+1} = -0.5$). With respect to the lap times under a SC, the lap time of the first car following the safety car will match its lap time, denoted as d_{SC} . As for the second car, its lap time will also match this time, except for the first lap of SC; in this case, the driver will catch up with the race leader reducing the time difference to half a second. Mathematically,

$$\mu_{n,SC}^A = \begin{cases} d_{SC} + p_{SC}^A \cdot \mathbb{1}_{\{x_n^A \neq 0\}} & \text{if } \xi(g_n, x_n^A, x_n^B) = 1 \\ d_{SC} + p_{SC}^B \cdot \mathbb{1}_{\{x_n^A \neq 0\}} - g_n + 0.5 & \text{otherwise,} \end{cases} \quad (20)$$

$$\mu_{n,SC}^B = \begin{cases} d_{SC} + p_{SC}^A \cdot \mathbb{1}_{\{x_n^B \neq 0\}} + g_n + 0.5 & \text{if } \xi(g_n, x_n^A, x_n^B) = 1 \\ d_{SC} + p_{SC}^B \cdot \mathbb{1}_{\{x_n^B \neq 0\}} & \text{otherwise.} \end{cases} \quad (21)$$

We can see from Equations (20) and (21) that the lap time of the leading car is d_{SC} plus the additional time of pitting, while the expression of the lap time of the second car (driver A if $\xi(g_n, x_n^A, x_n^B) = 0$ in Equation (20), and driver B if $\xi(g_n, x_n^A, x_n^B) = 1$ in Equation (21)) is such that $g_{n+1} = g_n + \mu_{n,SC}^A - \mu_{n,SC}^B$ with g_{n+1} in Equation (19).

For example, if a SC that lasts for 3 laps occurs at the end of lap $n = 15$ (i.e., $Y_{15}^{SC} = 3$), then g_{17} , g_{18} , and g_{19} will be ± 0.5 , while g_{16} takes the value of the time difference that the cars are having when the SC just happened.

4.2.3. Lap time under no yellow flag: We consider the uncertainty in the lap times by using separate random variables for the following two parts of the lap: between the start of the lap and the pit stop exit, and between the pit stop exit and the end of the lap. For the former, we use the random variable T_n^{DRS} to denote the lap time reduction of the following car due to the usage of DRS during lap n . We define a single random variable for this effect—and not one for each vehicle—since, at most, there will be one car using the DRS at each time. For the latter lap segment, we define two random variables, $Z_n^{(1)}$ and $Z_n^{(2)}$, that represent uncertainty in the interaction between the cars. The reason for using separate random variables for the two lap segments being considered, is that the time gap between cars could be very different between these (see the discussion before Equation (14)), and therefore the dynamics between the cars are also different. Also, we assume that the DRS can only be used in the former lap segment. We only show the lap time functions for car A , since those for B are analogous. The lap time for driver A is expressed as:

$$\mu_{n,0}^A(s_n, x_n^A, x_n^B, T_n^{DRS}, Z_n^{(1)}, Z_n^{(2)}) = \gamma_n^A + \eta_n'^A(\epsilon'_n) - T_n^{DRS} \cdot \mathbb{1}_{\{g_n \in [0,1], y_n^{DRS}=0\}}, \quad (22)$$

where γ^A denotes the *ghost lap time function* which is the same as the one defined for the deterministic case in Equation (13), and ϵ'_n denotes the time gap between cars after the pit stop exit (in the case when there is no yellow flag). This is computed by:

$$\epsilon'_n = g_n + p_0^A \cdot \mathbb{1}_{\{x_n^A \neq 0\}} - T_n^{DRS} \cdot \mathbb{1}_{\{g_n \in [0,1], y^{DRS}=0\}} - (p_0^B \cdot \mathbb{1}_{\{x_n^B \neq 0\}} - T_n^{DRS} \cdot \mathbb{1}_{\{g_n \in [-1,0], y^{DRS}=0\}}) \quad (23)$$

Finally, the *interaction function* in the stochastic case for car A , $\eta_n'^A$, is written as the following function:

$$\eta_n'^A(\epsilon'_n) = \begin{cases} \exp(-\lambda \cdot |\epsilon'_n|) \cdot Z_n^{(1)} & \text{if } \epsilon'_n < 0 \\ \exp(-\lambda \cdot |\epsilon'_n|) \cdot Z_n^{(2)} & \text{otherwise.} \end{cases} \quad (24)$$

For driver B , the *interaction function* $\eta_n'^B$ is defined in the same way as Equation (24) except that the condition in the RHS is changed between the two cases.

Putting together Equations (17), (20), (21), and (22) from the three cases, we can write the lap time function for driver $i \in \{A, B\}$ as:

$$\mu_n^i(s_n, x_n^A, x_n^B, T_n^{DRS}, Z_n^{(1)}, Z_n^{(2)}) = \begin{cases} \mu_{n,0}^i & \text{if } y_n^{VSC} + y_n^{SC} = 0 \\ \mu_{n,VSC}^i & \text{if } y_n^{VSC} > 0 \\ \mu_{n,SC}^i & \text{if } y_n^{SC} > 0, \end{cases} \quad (25)$$

where the arguments of the functions of the RHS have been omitted for simplicity. The model can also be extended to consider other racing aspects; for example, we can enforce a minimum time gap, $\Delta > 0$, between drivers in a point of the circuit. (This could enable modeling the fact that there are some parts of the circuit in which cars can not be arbitrarily close to each other.) This extension is described in Appendix C.

4.3. Stochastic Dynamic Programming

For ease of notation, let us define the random vector $W_n := (Y_n^{VSC}, Y_n^{SC}, T_n^{DRS}, Z_n^{(1)}, Z_n^{(2)})$. Then, the new of state variable (s_{n+1}) as a function of the previous state (s_n) , the decisions $(x_n^A$ and $x_n^B)$, and the random vector (W_n) , can be expressed as:

$$t_{n+1}^A = t_n^A \cdot \mathbb{1}_{\{x_n^A=0\}} + x_n^A \cdot \mathbb{1}_{\{x_n^A \neq 0\}} \quad (26)$$

$$w_{n+1}^A = \min\{w_n^A \cdot \mathbb{1}_{\{x_n^A=0\}} + \mathbb{1}_{\{Y_n^{VSC}+Y_n^{SC}=0\}}, u_{t_n} + 1\} \quad (27)$$

$$m_{n+1}^A = \max\{m_n^A, \mathbb{1}_{\{x_n^A \neq 0, x_n^A \neq t_n^A\}}\} \quad (28)$$

$$g_{n+1} = g_n + \mu_n'^A(s_n, x_n^A, x_n^B, W_n) - \mu_n'^B(s_n, x_n^A, x_n^B, W_n) \quad (29)$$

$$y_{n+1}^{VSC} = Y_n^{VSC} \quad (30)$$

$$y_{n+1}^{SC} = Y_n^{SC} \quad (31)$$

$$y_{n+1}^{DRS} = \max\{y_n^{DRS} - 1, 0\} \cdot \mathbb{1}_{\{Y_n^{VSC}+Y_n^{SC}=0\}} + \sum_{s \in \{VSC, SC\}} (Y_n^s + k^s) \cdot \mathbb{1}_{\{Y_n^s > 0\}}, \quad (32)$$

and similarly for t_{n+1}^B , w_{n+1}^B , and m_{n+1}^B . Note that tires do not wear under a yellow flag event (see Equation (27)). This is because of the significant reduction of speed during these events.

The objective of each driver is to maximize the time gap with respect to the other driver at the end of the race. Let us define the event $H^i = \{w_{N+1}^i \leq u_{t_{i+1}}\} \wedge \{m_{N+1}^i = 1\}$ for $i \in \{A, B\}$. The occurrence of the event H^i represents that driver i finishes the race adequately without using tires longer than their lifespans and using at least two different tire compounds. Consequently, the cost function at the border can be written as:

$$V_{N+1}^{I*}(s_{N+1}) = \begin{cases} g_{N+1} & \text{if } H^A \text{ and } H^B \\ -\infty & \text{if } H^A \text{ and not } H^B \\ +\infty & \text{if not } H^A \text{ and } H^B \\ 0 & \text{otherwise.} \end{cases} \quad (33)$$

Then, the Bellman Equation can be expressed as:

$$V'_n(s_n, x_n^A, x_n^B) = \mathbb{E}_{W_n} [V_{n+1}^*(s_{n+1})]. \quad (34)$$

For $n = 1, \dots, N - 1$

$$V_n^*(s_n) = \begin{cases} \min_{a \in \mathcal{T}_0} V'_n(s_n, a, x_n^B(s_n, a)) & \text{if } g_n < 0 \\ \max_{b \in \mathcal{T}_0} V'_n(s_n, x_n^A(s_n, b), b) & \text{if } g_n > 0 \end{cases}, \quad (35)$$

where

$$x_n^A(s_n, b) = \arg \min_{a \in \mathcal{T}_0} V'_n(s_n, a, b), \quad (36)$$

$$x_n^B(s_n, a) = \arg \max_{b \in \mathcal{T}_0} V'_n(s_n, a, b). \quad (37)$$

As in the deterministic case, we solve the simultaneous game for the first stage, in which both drivers decide the tire compound with which to start the race. For this, we define the matrix $U' \in \mathbb{R}^{|\mathcal{T}| \times |\mathcal{T}|}$ such that $U'_{t_1^A, t_1^B} = V_1^*((t_1^A, 0, 0, t_1^B, 0, 0, g_1, 0, 0, 2))$, for $t_1^A, t_1^B \in \mathcal{T}$. Note that we are considering that $y_1^{DRS} = 2$, which means that the DRS is going to be enabled in the third lap (as long as there are no yellow flag events in the first two laps). Then, the mixed strategies for the first stage are found by solving Optimization Problems (46) and (47) but with U' instead of U .

We denote by $\mathcal{S}_n$ the set of all states at lap n . It is important to note that, given that the leader at each stage is state dependent, a profile strategy of a player is either: (i) a function that only depends on the state, in which case the player is the leader, or (ii) a response function that depends on the state and on the other player's strategy, in which case the player corresponds to the follower. Thus, a profile strategy for a players A and B can be written as:

$$\begin{aligned} & \{x_n'^A(s_n)\}_{s_n \in \mathcal{S}_n^-, n \in \{1, \dots, N\}}, \{x_n'^A(s_n, b)\}_{s_n \in \mathcal{S}_n^+, b \in \mathcal{T}_0, n \in \{1, \dots, N\}}, \pi'^A \\ & \{x_n'^B(s_n)\}_{s_n \in \mathcal{S}_n^+, n \in \{1, \dots, N\}}, \{x_n'^B(s_n, a)\}_{s_n \in \mathcal{S}_n^-, a \in \mathcal{T}_0, n \in \{1, \dots, N\}}, \pi'^B, \end{aligned} \quad (38)$$

where $\mathcal{S}_n^-$ ($\mathcal{S}_n^+$) is the set of states of lap n where $g_n < 0$ (> 0). Namely, $\mathcal{S}_n^- = \{s_n \in \mathcal{S}_n | g_n < 0\}$ and $\mathcal{S}_n^+ = \{s_n \in \mathcal{S}_n | g_n > 0\}$.

A summary of the Stochastic Dynamic Programming is provided in Algorithm 2.

Algorithm 2 Stochastic Dynamic Programming Algorithm

- 1: **Input:** Set of states $\{\mathcal{S}_n\}_{n \in \{1, \dots, N+1\}}$. Starting time gap g_1
 - 2: **Output:** Decisions $\{x_n^{A*}(s_n)\}_{s_n \in \mathcal{S}_n^-, n \in \{1, \dots, N\}}, \{x_n^{A*}(s_n, b)\}_{s_n \in \mathcal{S}_n^+, b \in \mathcal{T}_0, n \in \{1, \dots, N\}},$
 $\{x_n^{B*}(s_n)\}_{s_n \in \mathcal{S}_n^+, n \in \{1, \dots, N\}}, \{x_n^{B*}(s_n, a)\}_{s_n \in \mathcal{S}_n^-, a \in \mathcal{T}_0, n \in \{1, \dots, N\}}, \pi^{A*}$ and π^{B*} .
 - 3: **Foreach** $s_n \in \mathcal{S}_{N+1}$
 - 4: Compute $V_{N+1}'(s_n)$ using Equation (33).
 - 5: **End For**
 - 6: **Set** $n \leftarrow N$
 - 7: **While** $n \geq 1$ **do**
 - 8: **Foreach** $s_n \in \mathcal{S}_n$
 - 9: Compute $V_n'(s_n, a, b)$ from Equation (34) for all $a, b \in \mathcal{T}_0$.
 - 10: **If** $g_n < 0$ **Then**
 - 11: Compute $x_n^{B*}(s_n, a)$ with Equation (37) for all $a \in \mathcal{T}_0$.
 - 12: $x_n^{A*}(s_n) \leftarrow \arg \max_{a \in \mathcal{T}_0} V_n'(s_n, a, x_n^{B*}(s_n, a)).$
 - 13: $V_n'(s_n) \leftarrow V_n'(s_n, x_n^{A*}(s_n), x_n^{B*}(s_n, x_n^{A*}(s_n))).$
 - 14: **Else**
 - 15: Compute $x_n^{A*}(s_n, b)$ with Equation (36) for all $b \in \mathcal{T}_0$.
 - 16: $x_n^{B*}(s_n) \leftarrow \arg \max_{b \in \mathcal{T}_0} V_n'(s_n, x_n^{A*}(s_n, b), b).$
 - 17: $V_n'(s_n) \leftarrow V_n'(s_n, x_n^{A*}(s_n, x_n^{B*}(s_n)), x_n^{B*}(s_n)).$
 - 18: **End If**
 - 19: **End For**
 - 20: $n \leftarrow n - 1$
 - 21: **End While**
 - 22: Compute $U' \in \mathbb{R}^{|\mathcal{T}| \times |\mathcal{T}|}$ such that $U'_{t_1^A, t_1^B} = V_1'((t_1^A, 0, 0, t_1^B, 0, 0, g_1, 0, 0, 2))$ for $t_1^A, t_1^B \in \mathcal{T}$
 - 23: Get π^{A*} and π^{B*} by solving Optimization Problems (46) and (47) with U' .
-

We define an equilibrium of the feedback Stackelberg game for a fixed starting tire compounds $t_1^A, t_1^B \in \mathcal{T}$ (encoded in s_1 , see Equation (16)).

DEFINITION 1. A *feedback Stackelberg equilibrium*, given an initial state s_1 , is a pair of strategies $\left(\{x_n^{A*}(s_n)\}_{s_n \in \mathcal{S}_n^-, n \in \{1, \dots, N\}}, \{x_n^{A*}(s_n, b)\}_{s_n \in \mathcal{S}_n^+, b \in \mathcal{T}_0, n \in \{1, \dots, N\}}\right)$ and $\left(\{x_n^{B*}(s_n)\}_{s_n \in \mathcal{S}_n^+, n \in \{1, \dots, N\}}, \{x_n^{B*}(s_n, a)\}_{s_n \in \mathcal{S}_n^-, a \in \mathcal{T}_0, n \in \{1, \dots, N\}}\right)$ such that for all $n \in \{1, \dots, N\}$:

1. If $g_n < 0$

$$V'_n(s_n, x_n^{A*}(s_n), x_n^{B*}(s_n, x_n^{A*}(s_n))) \leq V'_n(s_n, x_n'^A(s_n), x_n^{B*}(s_n, x_n'^A(s_n))) \quad (39)$$

$$V'_n(s_n, x_n^{A*}(s_n), x_n^{B*}(s_n, x_n^{A*}(s_n))) \geq V'_n(s_n, x_n^{A*}(s_n), x_n'^B(s_n, x_n^{A*}(s_n))). \quad (40)$$

2. If $g_n > 0$

$$V'_n(s_n, x_n^{A*}(s_n, x_n^{A*}(s_n)), x_n^{B*}(s_n)) \leq V'_n(s_n, x_n'^A(s_n, x_n^{B*}(s_n)), x_n^{B*}(s_n)) \quad (41)$$

$$V'_n(s_n, x_n^{A*}(s_n, x_n^{B*}(s_n)), x_n^{B*}(s_n)) \geq V'_n(s_n, x_n^{A*}(s_n, x_n'^B(s_n)), x_n'^B(s_n)), \quad (42)$$

for any $\left(\{x_n'^A(s_n)\}_{s_n \in \mathcal{S}_n^-, n \in \{1, \dots, N\}}, \{x_n'^A(s_n, b)\}_{s_n \in \mathcal{S}_n^+, b \in \mathcal{T}_0, n \in \{1, \dots, N\}} \right)$ and $\left(\{x_n'^B(s_n)\}_{s_n \in \mathcal{S}_n^+, n \in \{1, \dots, N\}}, \{x_n'^B(s_n, a)\}_{s_n \in \mathcal{S}_n^-, a \in \mathcal{T}_0, n \in \{1, \dots, N\}} \right)$.

We next show that the solution of Algorithm 2 for every state of the first lap, i.e., for all possible combinations of starting tire compounds, is a *feedback Stackelberg equilibrium*.

THEOREM 1. *For each pair of starting tire compound $(t_1^A, t_1^B) \in \mathcal{T}^2$, the output of Algorithm 2 results in a feedback Stackelberg equilibrium.*

Proof. See Appendix D. $\square$

We next define the equilibrium concept for the entire game, that is, including the simultaneous decisions on the starting tire compounds. We use the term *Nash-feedback Stackelberg* to denote this game since the initial tire decisions are simultaneous.

DEFINITION 2. A *Nash-feedback Stackelberg equilibrium* is a pair of strategies $\left(\{x_n^{A*}(s_n)\}_{s_n \in \mathcal{S}_n^-, n \in \{1, \dots, N\}}, \{x_n^{A*}(s_n, b)\}_{s_n \in \mathcal{S}_n^+, b \in \mathcal{T}_0, n \in \{1, \dots, N\}}, \pi^{A*} \right)$ and $\left(\{x_n^{B*}(s_n)\}_{s_n \in \mathcal{S}_n^+, n \in \{1, \dots, N\}}, \{x_n^{B*}(s_n, a)\}_{s_n \in \mathcal{S}_n^-, a \in \mathcal{T}_0, n \in \{1, \dots, N\}}, \pi^{B*} \right)$ if:

1. For every $(t_1^A, t_1^B) \in \mathcal{T}^2$, the part of the strategy that relates to the pit stop strategies during the race, i.e., $\left(\{x_n^{A*}(s_n)\}_{s_n \in \mathcal{S}_n^-, n \in \{1, \dots, N\}}, \{x_n^{A*}(s_n, b)\}_{s_n \in \mathcal{S}_n^+, b \in \mathcal{T}_0, n \in \{1, \dots, N\}} \right)$ and $\left(\{x_n^{B*}(s_n)\}_{s_n \in \mathcal{S}_n^+, n \in \{1, \dots, N\}}, \{x_n^{B*}(s_n, a)\}_{s_n \in \mathcal{S}_n^-, a \in \mathcal{T}_0, n \in \{1, \dots, N\}} \right)$, is a feedback Stackelberg equilibrium.
2. The part of the strategy that relates to the starting tire compound component strategy of both players, i.e., (π^{A*}, π^{B*}) , is a mixed Nash equilibrium of the simultaneous zero-sum game with action space $\mathcal{T}$ for both players and payoff function $-V_1((t_1, 0, 0, t_2, 0, 0, g_1, 0, 0, 2))$ for player A.

THEOREM 2. *The solution obtained from Algorithm 2 is a Nash-feedback Stackelberg equilibrium.*

Proof. The proofs follows directly from Line 22 of Algorithm 2, and use Theorem 1 to show the first condition of Definition 2. $\square$

Before going to the numerical results section, we introduce an alternative objective function in which players maximize the odds of winning the race. One argument in favour of using this as the players' objective is that they only care about winning, and not the final time difference to their opponent. However, it could also be argued that it is not just a matter of winning or loosing, but also there is a brand image that could be damaged by loosing by a large margin.

4.4. Maximize Probability of Winning

In this subsection we analyze the game setting in which both players' objectives are to maximize the probability of winning the race. We consider the same stochastic dynamic programming introduced before, but use the following utility function in the border condition (instead of Equation (33)):

$$V''_{N+1}(s_{N+1}) = \begin{cases} \mathbb{1}_{\{g_{N+1}>0\}} - \mathbb{1}_{\{g_{N+1}<0\}} & \text{if } H^A \text{ and } H^B \\ -1 & \text{if } H^A \text{ and not } H^B \\ 1 & \text{if not } H^A \text{ and } H^B \\ 0 & \text{otherwise.} \end{cases} \quad (43)$$

The equilibrium in this case can be defined in a similar manner as previously explained (see Definitions 1 and 2). It is worth noting that the equilibrium existence results can also be extended with minimal modifications. This is formally expressed in the following corollary:

COROLLARY 1. *The solution obtained from Algorithm 2 when replacing Equation (33) with Equation (43) is a Nash-feedback Stackelberg equilibrium for the game in which players maximize the odds of winning.*

Proof. The proof follows the same argument used above for Theorem 2. $\square$

The *Nash-feedback Stackelberg equilibrium* in this case can also be obtained with Algorithm 2 using Equation (43) as the border condition. Interestingly, the winning odds for each driver can be obtained directly from the equilibrium. Let us denote by $U'' \in \mathbb{R}^{|\mathcal{T}| \times |\mathcal{T}|}$ the drivers' expected payoff at the start of the race (similar to U'). Also, the mixed equilibrium at the start of the race under this context is denoted by $(\pi^{A'*}, \pi^{B'*})$.

PROPOSITION 1. *The probability that player A wins the race is equal to $(1 - (\pi^{A'*})^T U'' \pi^{B'*})/2$.*

Proof. See Appendix E. $\square$

Note that draws are counted as one half in the winning probability expression in Proposition 1. In what follows, we consider that in case of a draw, the winner is decided by a fair coin toss.

5. Numerical Results

We solve an instance of the problem described with the following parameters: $N = 52$, $u_1 = 30$, $u_2 = 40$, $u_3 = 50$, $d_0^A = 97.22$, $d_0^B = 97.24$, $p_0^A = 20.2$, $p_0^B = 20.0$, $\lambda = 2.0$, $h = 0.02$, $l^{VSC} = 2$, $l^{SC} = 3$, $r^{VSC} = 0.3$, $r^{SC} = 0.3$, $k^{VSC} = 2$, $k^{SC} = 2$, $d_{VSC} = 170.0$, $d_{SC} = 200.0$, $p_{VSC}^A = 10.0$, $p_{VSC}^B = 10.0$, $p_{SC}^A = 10.0$, $p_{SC}^B = 10.0$, $g_0 = -0.4$, and $\Delta = 0.4$. Regarding the random variables $Z^{(1)}$, $Z^{(2)}$, and T^{DRS} , we consider that: $Z^{(1)}$ takes the values 0.2, 0.4, and 0.6 with equal probability; $Z^{(2)}$ takes the values 0.5, 0.7, and 0.9 with equal probability; and T^{DRS} takes the values 0.1, 0.3, and 0.5 with the probabilities 0.2, 0.7, and 0.1, respectively. The tire-wear function used for soft, medium, and hard compound tires are described in Figure 2.

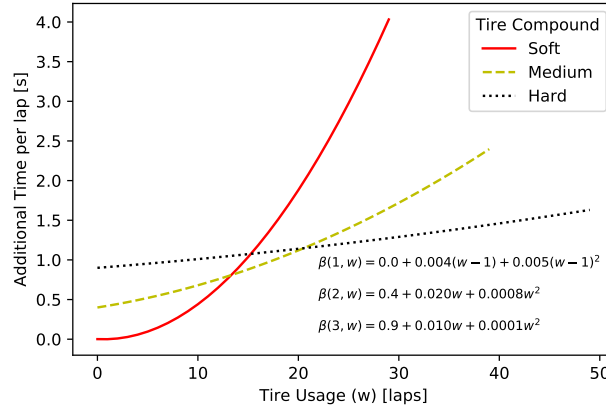


Figure 2 Additional lap times of tires with different usages compared to a set of new soft compound tires

In order to have a finite number of states, the time-gap state component is discretized. A grid between ± 35 [s] with a refinement of 0.04 [s] is used. The instance has hundreds of millions of states for laps greater than 15. Table 5 in Appendix F provides the exact number of states and solving time for some laps. Instances are solved on an HPC using 15

CPUs Intel of 2.10GHz with 20Gb RAM. Algorithm 2 is implemented in a parallel code in C.

5.1. Strategic players maximize time-gap

We apply Algorithm 2 to obtain the Nash-feedback Stackelberg equilibrium of the game. The payoff matrix of the tire' decisions at the beginning of the race (U') is shown in Table 1. Recall that these values represent the expected time difference at the end of the race where a negative value indicates A is finished ahead of B . The mixed strategies of the initial tire compounds for player A (B) are given in the right column (bottom row) of Table 1. We observe that both players randomize the election between the soft and medium compounds. Indeed, we see in Table 1 that the hard compound strategy is dominated by the soft compound for both players.

		Player B				
		Tire	Soft	Medium	Hard	π^{A*}
Player A	Soft	-0.11	-1.32	-2.95		0.41
	Medium	-0.93	-0.08	-2.91		0.59
	Hard	1.07	0.76	-0.08		0.00
		π^{B*}	0.60	0.40	0.00	

Table 1 Expected players' payoff at the beginning of the race, U' ; and mixed Nash Equilibrium, π^{A*} , π^{B*}

Once Algorithm 2 is run, we simulate 10,000 race scenarios with the parameters described at the beginning of Section 5, and the players' decisions from the Nash-feedback Stackelberg equilibrium. The first row of Table 2 shows various statistics of the simulations. Since driver A is two hundredths of a second per lap faster than B (although B is two tenths of a second faster in pitting), A wins in 66.01% of the scenarios.

Strategic Player	Objective Strat. Player	A wins [%]	Avg. g_{N+1} [s]	Sd. g_{N+1} [s]	Avg. g_{N+1} if		π^{A*}	π^{B*}
					A wins	B wins		
Both	Time Gap	66.01	-0.58	2.00	-1.42	1.03	(0.41, 0.59, 0.00)	(0.60, 0.40, 0.00)
Both	Win Prob.	66.53	-0.75	5.17	-2.31	2.35	(0.40, 0.60, 0.00)	(0.61, 0.39, 0.00)
A	Time Gap	84.09	-1.04	2.02	-1.59	1.82	(1.00, 0.00, 0.00)	(0.50, 0.50, 0.00)
B	Time Gap	50.37	-0.03	0.59	-0.59	0.53	(0.00, 1.00, 0.00)	(0.00, 1.00, 0.00)

Table 2 Distribution of wins, final time-gap mean and standard deviation, mean final time-gap conditioned to each winning player, and tire compound strategy with which to start the race (lap 1).

Figure 3 shows the players' strategies and the partial time difference (g_n) in two scenarios: the left panel shows a race with no yellow flags, while the right panel shows a race with a VSC in laps 31 and 32. In both cases shown in Figure 3, A (B) starts with the soft (medium) compound tire; thus, A plans for a short stint at the beginning followed by a longer stint, while B plans for the opposite strategy. With no yellow flag event, player A wins the race by slightly more than a second. However, the VSC of the scenario shown in the right panel of Figure 3 favors player B , who leverages by pitting during the VSC, losing less time than stopping during a regular lap. As a result, player B wins by 9 seconds in this case. Naturally, there are other scenarios in which a yellow flag benefits player A . (See Appendix G.)

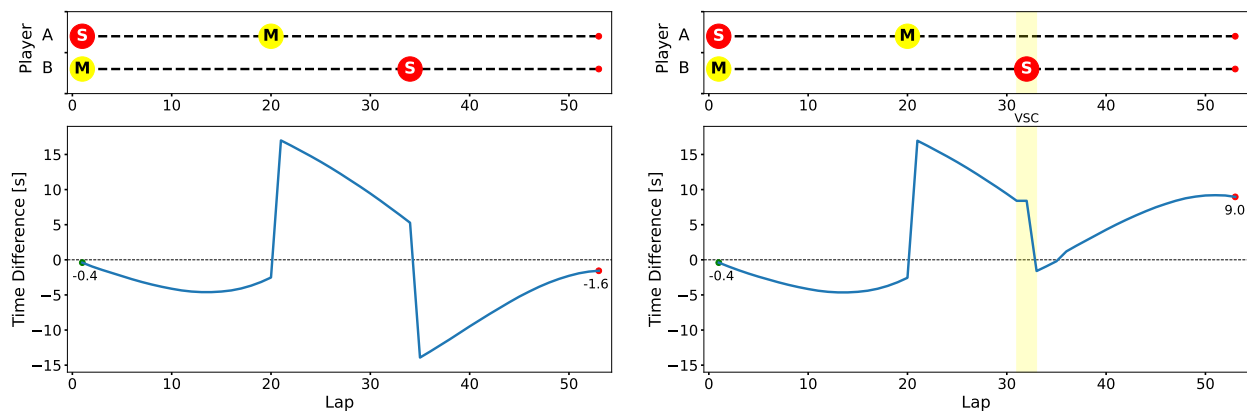


Figure 3 Race strategies of players (Top), partial race time difference (Bottom), scenario with no yellow flag (Left), and scenario with a VSC in laps 31 and 32 (Right).

5.2. Maximization of wins

The same instance as before is analyzed, except that players' objective is to maximize the probability of winning the race (see Section 4.4). In the third column of Table 2, it can be observed that A wins 66.53% of the time, similarly to the setting when both players maximize the time gap. However, in this case we observe that the standard deviation of the time-gap at the end of the race, in the fifth column of Table 2 (5.17 [s]), is significantly higher in this case compared to the case when players maximize time gap (2.00 [s]). The reason for this difference is illustrated in the next race scenario. Figure 4 shows a scenario with a SC in lap 17 in which both drivers change tires. Player B puts a soft compound tire which will not last until the end of the race, requiring an additional future stop. When players maximize the odds of winning (see the right panel of Figure 4), player B extends the

stint of this tire as much as possible, waiting for a yellow flag. Since no yellow flag occurs after lap 20, player *B* loses by more than 30 seconds (see the right panel of Figure 4). Note that by delaying the pit stop, player *B* manages to be ahead of *A* between laps 37 and 39 (unlike the case when players maximize time gap in the left panel of Figure 4); therefore, if there would be a yellow flag during these laps, both cars would stop for a Soft compound with *B* leading the race (see Appendix H for concrete example). By contrast, when players maximize the time-gap to their opponent, player *B* stops on lap 37, still losing at the end of the race, but only by 2.9 seconds (see the left panel of Figure 4). Then, if player *B* cares only about winning (and not about the time difference), he will take this chance despite losing by a longer time gap if the desired yellow flag event does not occur. This is similar to soccer, when getting to the end of the game a team is losing by one goal, its goal keeper goes in the opponents' area in case there is a corner kick.

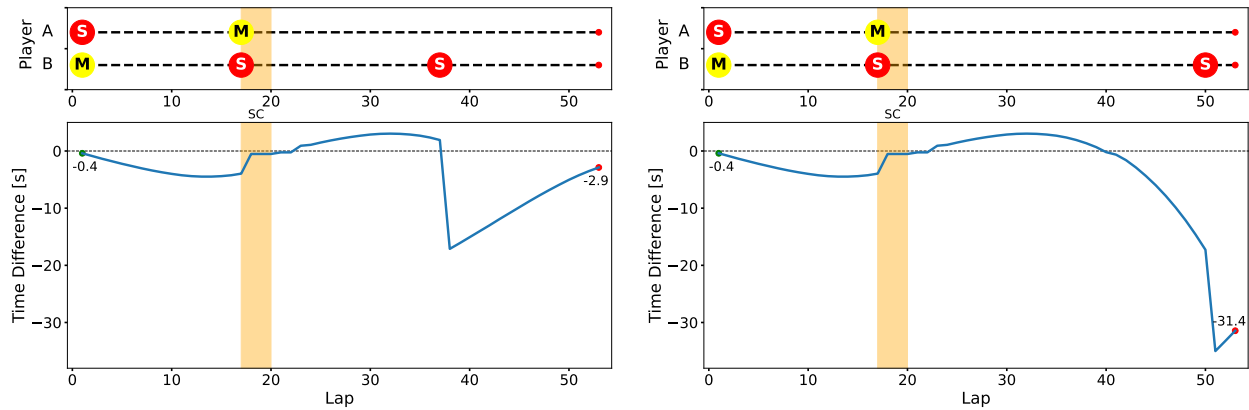


Figure 4 Race strategies of players (Top), partial race time difference (Bottom), players maximize the time gap (Left), players maximize the probability of winning (Right).

In addition we check (empirically) if Proposition 1 holds in this case. On the one hand, the theoretical winning probability of *A* according to Proposition 1, i.e., $(1 - (\pi^{A'*})^T U'' \pi^{B'*})/2$, equals to 67.16%. On the other hand, the 95% confidence interval around the simulated winning probability of *A* (66.53%) is [65.61%, 67.45%]. Thus, the theoretical winning probability for *A* falls inside the confidence interval. The values of the mixed equilibrium strategies of the first stage ($\pi^{A'*}$ and $\pi^{B'*}$) and the zero-sum matrix payoff (U'') are provided in Appendix I.

5.3. Ignoring competition

We analyze the same instance as in Section 5.1, except that player B ignores competition, and player A is aware of this. While the objective of player A is still to maximize the time-gap, the goal of player B is to minimize the total race time. In particular, player B considers the uncertainty of yellow flags, but ignores competitive effects, such as the *interaction function* of the lap time (see Equation (24)), and actions of player A . Then, the decisions of player B are obtained by solving a stochastic DP with the *ghost lap time function* as the cost-to-go function (see Equation (13)).

Figure 5 shows a race scenario where both players are strategic (left panel), and only player A is strategic (right panel). We see that when both players are strategic, their actions are equal during the whole race (i.e., both cars stop for the same tire compounds on the same laps). In this case, B happens to overtake A in two occasions (laps 14 and 48), while A overtakes B only once (lap 20); as a result, B wins the race (see left pane of Figure 5). However, when player B ignores competition, player A stops one lap before his opponent (in lap 19) performing an *undercut*⁸ maneuver. Although A is in second place by lap 19, his 20th lap is much faster than that of B , since A is with new tires. Hence, B loses the lead being behind of A by more than one second⁹ after his pit stop in lap 20 (i.e., the undercut is performed). From this point, player A maintains his race position until the end of the race (see right panel of Figure 5).

⁸ An undercut is a maneuver in which the car that is closely following the other stops a few laps before his opponent. The driver who is following is able to achieve faster lap times than his opponent (since the latter is using tires with more wear). Then, when the leader stops to change tires, he ends up behind the car that stopped earlier.

⁹ Note that if the time gap is more than a second, the follower can not activate the DRS, and therefore is harder for B to overtake A in this case (unlike the case in which both are strategic after lap 20, where A was also ahead of B but with a time gap less than a second).

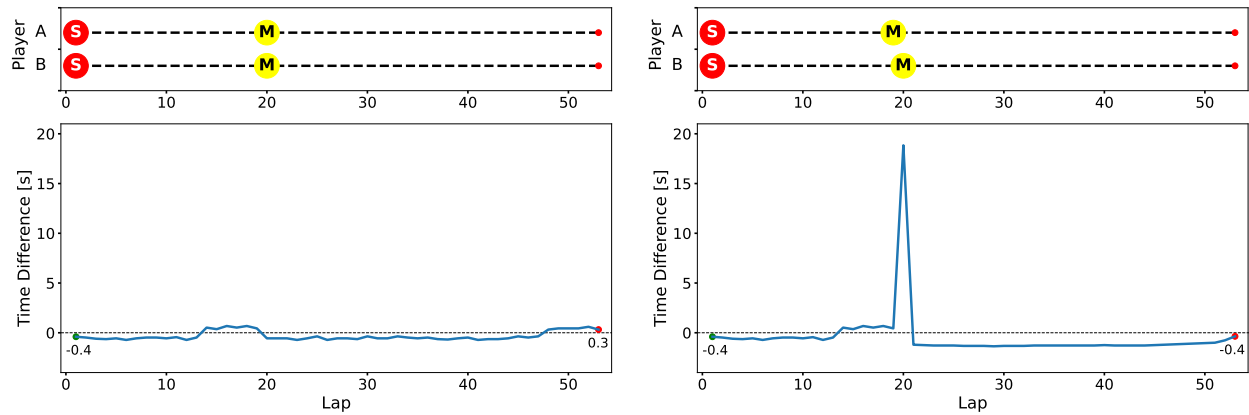


Figure 5 Race strategies of players (Top), partial race time difference (Bottom), players are strategic (Left), player *A* is strategic and *B* ignores competition (Right).

From the simulations, we observe (on the third row of Table 2 without considering headers) that player *A* increases his chances of winning by more than 15% now that *B* ignores competition.

5.4. Effect of yellow flags

We analyze the impact of yellow flags on the winning odds of drivers. The same instance as the one described in Section 5 is considered, except that the probability of yellow flag events is varied. More specifically, the probability of having a SC or VSC during the race is considered to be equal, taking values from 0% to 50%. Once running Algorithm 2 for each case, we run 10,000 simulated scenarios.

Figure 6 shows the winning probability of these simulations for the different yellow flag cases. We can see that the more likely yellow flags are to occur, the higher the benefit is to the driver with lower odds of winning, e.g., player *B* in this case. This is probably because of two reasons. First, a SC event allows the driver who is behind to close the time gap with the leader, somehow resetting the race, and thereby increasing his chances of winning. Second, as seen in the right panel of Figure 3, more yellow flags induce uncertainty in the race outcome; therefore, increasing the winning chances of the driver with the fewer odds.

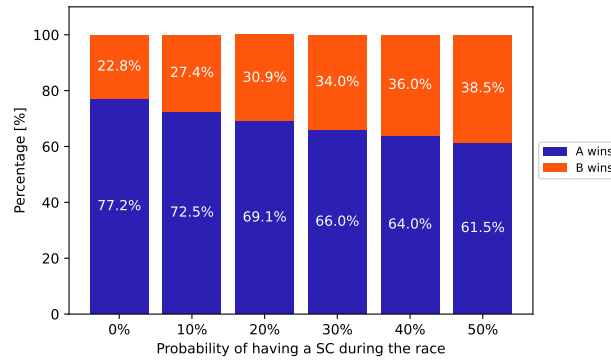


Figure 6 Winning distribution when the probability of a SC during the race goes from 0% to 50%. Probabilities of having a VSC or a SC are the same.

6. Conclusions and Future Directions

It has been shown that optimization of race strategies with competition between two drivers can be modeled as a feedback Stackelberg game using Stochastic Dynamic Programming. This allows accounting for race interaction events between drivers, and stochastic events such as yellow flag scenarios, lap times, etc. Once a feedback-Stackelberg equilibrium has been defined, we showed its existence and provided an algorithm to find this. In addition to the sequential pit stop decisions during the race, the model also accounts for the simultaneous initial tire compound decisions.

Despite the numerical instances of the problem have of the order of hundreds of millions of states per lap, these can be solved in a few hours. From these instances, we observe how different race scenarios affect players' decisions and the race outcome. We observe how drivers' different objective functions induce different race strategies. Indeed, when players maximize the probability of winning instead of the time-gap, their actions tend to be more risk taking. Observations from the numerical computations revealed that there is a significant advantage of being strategic when facing an opponent that ignores competition, in particular, the strategic player increases his winning odds by more 15% compared to the case when both are strategic. Finally, we observe that yellow flags tend to increase the chances of winning for the driver with the lower odds.

There are several research directions in which the model presented could be extended. One of these is to consider competition with more than two drivers. The latter is clearly not trivial since the size of the instance would considerably grow. Additionally, the model could be adapted to other applications beyond Formula 1 (or racing), as in various industries

or decision-making processes that need to allocate the timing of a process while facing interaction between other players; for example, industries that need to optimize a schedule for maintenance of equipment when all the firms' maintenance decisions affect each other. Also, the model could be useful for solving market share competition problems, such as for a television program that optimizes the allocation of ads in order to maximize rating; or in a political campaign where a "pit-stop" could be a new image increasing the candidate's chances to connect with multiple target audiences.

Acknowledgments

The authors gratefully acknowledge financial support from CONICYT PIA/BASAL AFB220003. This research was partially supported by the supercomputing infrastructure of the NLHPC (ECM-02).

References

- Anselma PG (2022) Optimal adaptive race strategy for a formula-e car. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering* 236(9):2185–2199.
- Başar T, Olsder GJ (1998) *Dynamic noncooperative game theory* (SIAM).
- Bekker J, Lotz W (2009) Planning formula one race strategies using discrete-event simulation. *Journal of the Operational Research Society* 60(7):952–961.
- Biswas S (2022) What is the difference between a safety car and a virtual safety car in f1? <https://www.essentiallysports.com/f1-news-what-is-the-difference-between-a-safety-car-and-a-virtual-safety-car-in-f1/>, (accessed March, 2023).
- Chain Bear (2017) Basics of f1 race strategy. https://youtu.be/wqf-dJyU_WA, accessed April, 2021.
- Choo CLW (2015) *Real-time decision making in motorsports: analytics for improving professional car race strategy*. Ph.D. thesis, Massachusetts Institute of Technology.
- F1 Strategy Blog (2019) Race simulator: Downloadable r program code. <https://f1strategyblog.wordpress.com/2019/05/10/race-simulator-downloadable-r-program-code/>, accessed December, 2022.
- Formula 1 (2021) Brilliant verstappen claims maiden title after victory in abu dhabi. <https://www.formula1.com/en/latest/article.brilliant-verstappen-claims-maiden-title-after-victory-in-abu-dhabi-season.22Arui4l6BKK2PUsJgKqpZ.html>, accessed December, 2022.
- Fortune (2015) How formula 1 teams use big data to win. <https://fortune.com/2015/11/12/big-data-formula-1-championship-race/>, (accessed December, 2022).
- Heilmeier A, Graf M, Betz J, Lienkamp M (2020a) Application of monte carlo methods to consider probabilistic effects in a race simulation for circuit motorsport. *Applied Sciences* 10(12):4229.
- Heilmeier A, Graf M, Lienkamp M (2018) A race simulation for strategy decisions in circuit motorsports. *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, 2986–2993 (IEEE).

- Heilmeier A, Thomaser A, Graf M, Betz J (2020b) Virtual strategy engineer: Using artificial neural networks for making race strategy decisions in circuit motorsport. *Applied Sciences* 10(21):7805.
- Heine OFC, Thraves C (2022) On the optimization of pit stop strategies via dynamic programming. *Central European Journal of Operations Research* 1–30.
- Li T, Sethi SP (2017) A review of dynamic stackelberg game models. *Discrete & Continuous Dynamical Systems-B* 22(1):125.
- Liu X, Fotouhi A, Auger DJ (2021) Formula-e race strategy development using distributed policy gradient reinforcement learning. *Knowledge-Based Systems* 216:106781.
- McLaren Racing Limited (2019) Formula one race strategy. <https://www.stem.org.uk/resources/elibrary/resource/25407/formula-one-race-strategy#&gid=undefined&pid=1>, accessed December, 2021.
- MotorSports (2022) Insider's guide: What are f1's tyre rules? <https://us.motorsport.com/f1/news/insiders-guide-what-are-f1s-tyre-rules/6885459/>, (accessed March, 2023).
- Nie Py, Lai My, Zhu Sj (2008) Dynamic feedback stackelberg games with alternating leaders. *Nonlinear Analysis: Real World Applications* 9(2):536–546.
- Phillips A (2014) Building a race simulator. <https://flmetrics.wordpress.com/2014/10/03/building-a-race-simulator/>, accessed December, 2022.
- Simaan M, Cruz JB (1973a) Additional aspects of the stackelberg strategy in nonzero-sum games. *Journal of Optimization Theory and Applications* 11(6):613–626.
- Simaan M, Cruz JB (1973b) On the stackelberg strategy in nonzero-sum games. *Journal of Optimization Theory and Applications* 11(5):533–555.
- The Sport Rush (2020) How does qualifying work in f1? qualifying system explained. <https://thesportsrush.com/f1-news-how-does-qualifying-work-in-f1-explained/>, accessed December, 2022.
- Tulabandhula T, Rudin C (2014) Tire changes, fresh air, and yellow flags: challenges in predictive analytics for professional racing. *Big data* 2(2):97–112.
- Von Stengel B (2002) Computing equilibria for two-person games. *Handbook of game theory with economic applications* 3:1723–1759.

Appendix A: Summary of Parameters

A summary of the notation of all the parameters is shown in Table 3. Similarly, a summary of the decisions, states, and random variables is provided in Table 4.

Notation	Description	Range
$\mathcal{T}$	Set of tires	$\{1, 2, 3\}$
N	Laps of the race	$\mathbb{Z}_+$
u_t	Lifespan of tire $t \in \mathcal{T}$	$\mathbb{Z}_+$
d_0^i	Lap time for driver $i \in \{A, B\}$ when using new soft compound tires at the beginning of the race (without pit stops, interactions, or DRS)	$[0, \infty)$
p_0^i	additional lap time for driver $i \in \{A, B\}$ due to a pit stop	$[0, \infty)$
λ	factor that penalizes the drivers' lap times because of being too close to each other	$[0, \infty)$
h	lap time reduction between two consecutive laps attributed to fuel consumption (making the car lighter)	$[0, \infty)$
l^{VSC}	laps that last a VSC	$\mathbb{Z}_+$
l^{SC}	laps that last a SC	$\mathbb{Z}_+$
q^{VSC}	probability of having a VSC during a lap	$[0, \infty)$
q^{SC}	probability of having a SC during a lap	$[0, \infty)$
r^{VSC}	probability of having at least one VSC during the race	$[0, \infty)$
r^{SC}	probability of having at least one SC during the race	$[0, \infty)$
k^{VSC}	number of laps needed to be raced after the end of a VSC to enable the DRS	$\mathbb{Z}_+$
k^{SC}	number of laps needed to be raced after the end of a SC to enable the DRS	$\mathbb{Z}_+$
d_{VSC}	Lap time during a VSC (without pit stop)	$[0, \infty)$
d_{SC}	Lap time during a SC (without pit stop)	$[0, \infty)$
p_{VSC}^i	additional lap time for driver $i \in \{A, B\}$ due to a pit stop under VSC	$[0, \infty)$
p_{SC}^i	additional lap time for driver $i \in \{A, B\}$ due to a pit stop under SC	$[0, \infty)$
t^{DRS}	Time reduction due to the use of the DRS in lap n (only in deterministic DP)	$\mathbb{R}$
$z_n^{(1)}$	Time reduction factor to the car in front due to interactions with the car behind in lap n (only in deterministic DP)	$[0, \infty)$
$z_n^{(2)}$	Time reduction factor to the car in the back due to interactions with the leading car in lap n (only in deterministic DP)	$[0, \infty)$
g_0	Time difference at the start of the race. Negative (positive) if A (B) starts first.	$\mathbb{R}$
Δ	Minimum time difference between drivers at the pit stop exit (see Appendix C)	$[0, \infty)$

Table 3 Parameters of the problem

Notation	Description	Range	Type	Problem
x_n^A	Tire compound changed by driver A in the pit stop in lap n if $x_n^A \in \mathcal{T}$; no pit stop during lap n if $x_n^A = 0$	$\{0\} \cup \mathcal{T}$	Decision	DP,SDP
x_n^B	Tire compound changed by driver B in the pit stop in lap n if $x_n^B \in \mathcal{T}$; no pit stop during lap n if $x_n^B = 0$	$\{0\} \cup \mathcal{T}$	Decision	DP,SDP
t_n^A	Tire of driver A at the beginning of lap n	$\mathcal{T}$	State	DP,SDP
t_n^B	Tire of driver B at the beginning of lap n	$\mathcal{T}$	State	DP,SDP
w_n^A	Usage (in laps) of the tires of driver A at the beginning of lap n	$\{0, 1, \dots, u_t + 1\}$	State	DP,SDP
w_n^B	Usage (in laps) of the tires of driver B at the beginning of lap n	$\{0, 1, \dots, u_t + 1\}$	State	DP,SDP
m_n^A	Indicator equal to 1 if A has used at least two different tire compounds by the beginning of lap n , 0 otherwise	$\{0, 1\}$	State	DP,SDP
m_n^B	Indicator equal to 1 if B has used at least two different tire compounds by the beginning of lap n , 0 otherwise	$\{0, 1\}$	State	DP,SDP
g_n	Time difference between cars at the beginning of lap n . $g_n < 0$ if A is first	$\mathbb{R}$	State	DP,SDP
y_n^{VSC}	Remaining laps under VSC from the beginning of lap n	$\{0, 1, \dots, l^{VSC}\}$	State	SDP
y_n^{SC}	Remaining laps under SC from the beginning of lap n	$\{0, 1, \dots, l^{SC}\}$	State	SDP
y_n^{DRS}	Remaining laps to enable the DRS from the beginning of lap n	$\mathbb{Z}_+$	State	SDP
π^A	Vector with the probabilities that A chooses each type of tire at the start of the race	$[0, 1]^{ \mathcal{T} }$	Decision	DP,SDP
π^B	Vector with the probabilities that B chooses each type of tire at the start of the race	$[0, 1]^{ \mathcal{T} }$	Decision	DP,SDP
Y_n^{VSC}	Remaining laps under VSC in lap n	$\{0, \dots, l^{VSC}\}$	RV	SDP
Y_n^{SC}	Remaining laps under SC in lap n	$\{0, \dots, l^{SC}\}$	RV	SDP
T_n^{DRS}	Time reduction due to the use of the DRS in lap n	$\mathbb{R}$	RV	SDP
$Z_n^{(1)}$	Time reduction factor to the car in front due to interactions with the car behind in lap n	$[0, \infty)$	RV	SDP
$Z_n^{(2)}$	Time reduction factor to the car in the back due to interactions with the leading car in lap n	$[0, \infty)$	RV	SDP

Table 4 Decisions, states, and random variables (RV) of the problem

Appendix B: Finding Mixed Nash Equilibrium at the Start of the Race

Both players solve the following min-max game:

$$\begin{aligned}
& \max_{\pi^B} \min_{\pi^A} (\pi^B)^T U^T \pi^A \\
& \text{s.t.} \quad \sum_{i \in \mathcal{T}} \pi_i^B = 1 \\
& \quad \pi_i^B \geq 0 \quad \forall i \in \mathcal{T} \\
& \quad \sum_{i \in \mathcal{T}} \pi_i^A = 1 \\
& \quad \pi_i^A \geq 0 \quad \forall i \in \mathcal{T}.
\end{aligned} \tag{44}$$

Then, the dual of the inner minimization problem of Optimization Problem (44) is:

$$\begin{aligned}
& \max_{\rho} \rho \\
& \text{s.t.} \quad \rho - e_i^T U \pi^B \leq 0 \quad \forall i \in \mathcal{T},
\end{aligned} \tag{45}$$

where $e_i \in \mathbb{R}^{|\mathcal{T}|}$ is a vector with a 1 in the i^{th} component and zero elsewhere. Putting together Optimization Problems (44) and (45) we get:

$$\begin{aligned} \max_{\rho, \pi^B} \quad & \rho \\ \text{s.t.} \quad & \rho - e_i^T U \pi^B \leq 0 \quad \forall i \in \mathcal{T} \\ & \sum_{i \in \mathcal{T}} \pi_i^B = 1 \\ & \pi_i^B \geq 0 \quad \forall i \in \mathcal{T}. \end{aligned} \quad (46)$$

An analogous procedure can be done to obtain π^{A*} , by taking the dual of the optimization problem faced by player B in Equation (44) plus the simplex constraints of the mixing probabilities. This leads to the following optimization problem:

$$\begin{aligned} \min_{\rho, \pi^A} \quad & \rho \\ \text{s.t.} \quad & \rho - e_i^T U^T \pi^A \geq 0 \quad \forall i \in \mathcal{T} \\ & \sum_{i \in \mathcal{T}} \pi_i^A = 1 \\ & \pi_i^A \geq 0 \quad \forall i \in \mathcal{T}. \end{aligned} \quad (47)$$

Appendix C: Minimum time gap

In this section we provide the details of the changes needed on the model presented in Section 4 to enforce a minimum time gap between cars at a point of the circuit. The rationale for this is that circuits usually have sections where cars are almost enforced to follow a *race line*; therefore, imposing a minimum time gap between cars. In this case, we consider that cars must have a minimum time difference of $\Delta > 0$ after the pit stop exit. The two changes needed are the following: (i) First, the *interaction function* should be changed to:

$$\eta_n''^A(\epsilon'_n) = \begin{cases} \exp(-\lambda \cdot |\min\{-\Delta, \epsilon'_n\}|) \cdot Z_n^{(1)} & \text{if } \epsilon'_n < 0 \\ \exp(-\lambda \cdot |\max\{\Delta, \epsilon'_n\}|) \cdot Z_n^{(2)} & \text{otherwise.} \end{cases} \quad (48)$$

In the case of driver B , the *interaction function* $\eta_n''^B$ is defined in the same way as in Equation (48) except that the condition in the RHS is changed between the two cases.

(ii) The second change, is with respect to the lap time function when there is no yellow flag, which changes to:

$$\begin{aligned} \mu_{n,0}'^A(s_n, x_n^A, x_n^B, T_n^{DRS}, Z_n^{(1)}, Z_n^{(2)}) &= \gamma_n^A + \eta_n''^A(\epsilon'_n) - T_n^{DRS} \cdot \mathbb{1}_{\{g_n \in [0,1], y_n^{DRS}=0\}} + \max\{\Delta - \epsilon, 0\} \cdot \mathbb{1}_{\{\epsilon > 0\}}, \\ \mu_{n,0}'^B(s_n, x_n^A, x_n^B, T_n^{DRS}, Z_n^{(1)}, Z_n^{(2)}) &= \gamma_n^B + \eta_n''^B(-\epsilon'_n) - T_n^{DRS} \cdot \mathbb{1}_{\{g_n \in [0,1], y_n^{DRS}=0\}} + \max\{\Delta + \epsilon, 0\} \cdot \mathbb{1}_{\{\epsilon \leq 0\}}. \end{aligned}$$

Then the lap time function should be updated to

$$\mu_n''^i(s_n, x_n^A, x_n^B, T_n^{DRS}, Z_n^{(1)}, Z_n^{(2)}) = \begin{cases} \mu_{n,0}'^i & \text{if } y_n^{VSC} + y_n^{SC} = 0 \\ \mu_{n,VSC}^i & \text{if } y_n^{VSC} > 0 \\ \mu_{n,SC}^i & \text{if } y_n^{SC} > 0, \end{cases} \quad (49)$$

for $i \in \{A, B\}$, and Equation (29) should be updated so that it uses Equation (49) instead of Equation (25).

Appendix D: Proof of Theorem 1

The result is proven by using induction in n from the last lap to the first one. Before doing this, Definition 1 is generalized for the case in which the game is at an intermediate stage (i.e., lap) $j \in \{1, \dots, N\}$ for any arbitrary state $s_j \in \mathcal{S}_j$. Let us define by, with some abuse of notation, $\mathcal{S}_n(s_j)$ the set of reachable states from state s_j at stage n . That is to say, reachable in the sense that there is a non-zero probability of arriving at this state in lap n , considering all the possible combinations of actions of both players and realization random variables between laps j and $n - 1$, and given that we are at state s_j in lap $j \leq n$. Clearly, $\mathcal{S}_j(s_j) = \{s_j\}$. Similar as before, we also define the sets $\mathcal{S}_n^-(s_j) = \{s_n \in \mathcal{S}_n(s_j) | g_n < 0\}$ and $\mathcal{S}_n^+(s_j) = \{s_n \in \mathcal{S}_n(s_j) | g_n > 0\}$ for all $n \geq j$ (note that g_n is a component of the state s_n).

DEFINITION 3. A *feedback Stackelberg equilibrium* from a state s_j at a given stage $j \in \{1, \dots, N\}$, is a pair of strategies: $\left(\{x_n^{A*}(s_n)\}_{s_n \in \mathcal{S}_n^-(s_j), n \in \{j, \dots, N\}}, \{x_n^{A*}(s_n, b)\}_{s_n \in \mathcal{S}_n^+(s_j), b \in \mathcal{T}_0, n \in \{j, \dots, N\}} \right)$ and $\left(\{x_n^{B*}(s_n)\}_{s_n \in \mathcal{S}_n^+(s_j), n \in \{1, \dots, N\}}, \{x_n^{B*}(s_n, a)\}_{s_n \in \mathcal{S}_n^-(s_j), a \in \mathcal{T}_0, n \in \{j, \dots, N\}} \right)$ such that for all $n \in \{j, \dots, N\}$:

1. If $g_n < 0$

$$V'_n(s_n, x_n^{A*}(s_n), x_n^{B*}(s_n, x_n^{A*}(s_n))) \leq V'_n(s_n, x_n^{A'}(s_n), x_n^{B*}(s_n, x_n^{A'}(s_n))) \quad (50)$$

$$V'_n(s_n, x_n^{A*}(s_n), x_n^{B*}(s_n, x_n^{A*}(s_n))) \geq V'_n(s_n, x_n^{A*}(s_n), x_n^{B'}(s_n, x_n^{A*}(s_n))). \quad (51)$$

2. If $g_n > 0$

$$V'_n(s_n, x_n^{A*}(s_n, x_n^{A*}(s_n)), x_n^{B*}(s_n)) \leq V'_n(s_n, x_n^{A'}(s_n, x_n^{B*}(s_n)), x_n^{B*}(s_n)) \quad (52)$$

$$V'_n(s_n, x_n^{A*}(s_n, x_n^{B*}(s_n)), x_n^{B*}(s_n)) \geq V'_n(s_n, x_n^{A*}(s_n, x_n^{B'}(s_n)), x_n^{B'}(s_n)), \quad (53)$$

for any $\left(\{x_n^{A'}(s_n)\}_{s_n \in \mathcal{S}_n^-(s_j), n \in \{j, \dots, N\}}, \{x_n^{A'}(s_n, b)\}_{s_n \in \mathcal{S}_n^+(s_j), b \in \mathcal{T}_0, n \in \{j, \dots, N\}} \right)$ and $\left(\{x_n^{B'}(s_n)\}_{s_n \in \mathcal{S}_n^+(s_j), n \in \{j, \dots, N\}}, \{x_n^{B'}(s_n, a)\}_{s_n \in \mathcal{S}_n^-(s_j), a \in \mathcal{T}_0, n \in \{j, \dots, N\}} \right)$.

Note that in Definition 3 we are considering the game from stage j at a particular state s_j . Thus, this is also a feedback Stackelberg game where the players' strategies are involved from stage j to N . Clearly, Definition 3 matches with Definition 1 for the cases when s_j (from Definition 3) refers to the same initial state s_1 (from Definition 1).

Let us denote the output of Algorithm 2 as:

$$\begin{aligned} & \{x_n^{\diamond A}(s_n)\}_{s_n \in \mathcal{S}_n^-, n \in \{1, \dots, N\}}, \{x_n^{\diamond A}(s_n, b)\}_{s_n \in \mathcal{S}_n^+, b \in \mathcal{T}_0, n \in \{1, \dots, N\}} \\ & \{x_n^{\diamond B}(s_n)\}_{s_n \in \mathcal{S}_n^+, n \in \{1, \dots, N\}}, \{x_n^{\diamond B}(s_n, a)\}_{s_n \in \mathcal{S}_n^-, a \in \mathcal{T}_0, n \in \{1, \dots, N\}}, \end{aligned} \quad (54)$$

where we are not considering the mixed strategy of the first stage.

We show first the base case, i.e., the last lap $n = N$. Consider a fixed state $s_N \in \mathcal{S}_N$. Note that since there is only one stage left in this case, this is a classical Stackelberg game, in which the leader depends on the component g_N of the state s_N . We will treat the case where the component $g_N < 0$, i.e., A is first in the last lap. The other case is omitted since its analysis is analogous. Driver B has no incentive to change its strategy, indeed, for any pair of strategies $(x_N^{\diamond A}(s_N), \{x_N^{B'}(s_N, a)\}_{a \in \mathcal{T}_0})$ it happens that

$$\begin{aligned} V'_N(s_N, x_N^{\diamond A}(s_N), x_N^{\diamond B}(s_N, x_N^{\diamond A}(s_N))) &= \max_{b \in \mathcal{T}_0} V'_N(s_N, x_N^{\diamond A}(s_N), b) \\ &\geq V'_N(s_N, x_N^{\diamond A}(s_N), x_N^{B'}(s_N, x_N^{\diamond A}(s_N))), \end{aligned}$$

where the equality is because of Equation (37) used in Line 11 of Algorithm 2, and the inequality is because we are considering a particular action for B , $x'_N(s_N, x_N^{\diamond A}(s_N))$ in this case. Similarly player A has no incentive to change its strategy. Consider any pair of strategies $(x'_N(s_N), \{x_N^{\diamond B}(s_N, a)\}_{a \in \mathcal{T}_0})$. Then,

$$\begin{aligned} V'_N(s_N, x_N^{\diamond A}(s_N), x_N^{\diamond B}(s_N, x_N^{\diamond A}(s_N))) &= \min_{a \in \mathcal{T}_0} V'_N(s_N, a, x_N^{\diamond B}(s_N, a)) \\ &\leq V'_N(s_N, x'_N(s_N), x_N^{\diamond B}(s_N, x'_N(s_N))), \end{aligned}$$

where the equality is because of Line 12 of Algorithm 2, and the inequality is because we are considering a particular action for A , $x'_N(s_N)$.

Consider the hypothesis induction to be true for all stages $n \geq j+1$, for a fixed $j \in \{1, \dots, N-1\}$. (Note that j is used as the index of the induction.) In other words, for all states $s_n \in \mathcal{S}_n$ for $n \geq j+1$, the output of Algorithm 2 (see Equation (54)), satisfies Definition 3. Consider we are at stage j in a fixed state $s_j \in \mathcal{S}_j$. As in the base case, only the case where the component $g_j < 0$ is considered.

We first show that B has no incentive to change its strategy. Indeed, consider any pair of strategies where B deviates from the strategy of Equation (54), namely,

$$\begin{aligned} &\{x_n^{\diamond A}(s_n)\}_{s_n \in \mathcal{S}_n^-(s_j), n \in \{j, \dots, N\}}, \{x_n^{\diamond A}(s_n, b)\}_{s_n \in \mathcal{S}_n^+(s_j), b \in \mathcal{T}_0, n \in \{j, \dots, N\}} \\ &\{x_n^{\diamond B}(s_n)\}_{s_n \in \mathcal{S}_n^+(s_j), n \in \{j, \dots, N\}}, \{x_n^{\diamond B}(s_n, a)\}_{s_n \in \mathcal{S}_n^-(s_j), a \in \mathcal{T}_0, n \in \{j, \dots, N\}}. \end{aligned} \quad (55)$$

Consider the two following cases with respect to the strategy of player B :

- (i) If the strategy of B of Equation (55) matches the one obtained by Algorithm 2 on stage j , i.e., $x'^B(s_j, a) = x^{\diamond B}(s_j, a)$ for all $a \in \mathcal{T}$ (note we are explicitly denoting B as the follower at stage j since $g_j < 0$), then it clearly holds that:

$$V'_j(s_j, x_j^{\diamond A}(s_j), x_j^{\diamond B}(s_j, x_j^{\diamond A}(s_j))) = V'_j(s_j, x_j^{\diamond A}(s_j), x_j'^B(s_j, x_j^{\diamond A}(s_j))),$$

and therefore Equation (51) holds. Then, for all other states, $s_n \in \mathcal{S}_n(s_j)$ for $n > j$, Equations (51), and (53) hold because of the hypothesis induction. Then, player B has no incentive to deviate in this case.

- (ii) If the strategy of B in Equation (55) does not coincide with the one obtained from Algorithm 2 at stage j , i.e., $x'^B(s_j, a) \neq x^{\diamond B}(s_j, a)$ for some $a \in \mathcal{T}_0$, then:

$$\begin{aligned} V'_j(s_j, x_j^{\diamond A}(s_j), x_j^{\diamond B}(s_j, x_j^{\diamond A}(s_j))) &= \max_{b \in \mathcal{T}_0} V'_j(s_j, x_j^{\diamond A}(s_j), b) \\ &\geq V'_j(s_j, x_j^{\diamond A}(s_j), x_j'^B(s_j, x_j^{\diamond A}(s_j))), \end{aligned}$$

where the equality is because of Equation (37) used in Line 11 of Algorithm 2, and the inequality is because we are considering a particular action for B , $x_j'^B(s_j, x_j^{\diamond A}(s_j))$ in this case. I.e., B has no incentive to deviate at stage j (at state s_j).

We now show that A has no incentive to deviate from the output strategy of Algorithm 2 (Equation (54)). Consider any pair of strategies

$$\begin{aligned} &\{x_n'^A(s_n)\}_{s_n \in \mathcal{S}_n^-(s_j), n \in \{j, \dots, N\}}, \{x_n'^A(s_n, b)\}_{s_n \in \mathcal{S}_n^+(s_j), b \in \mathcal{T}_0, n \in \{j, \dots, N\}} \\ &\{x_n^{\diamond B}(s_n)\}_{s_n \in \mathcal{S}_n^+(s_j), n \in \{j, \dots, N\}}, \{x_n^{\diamond B}(s_n, a)\}_{s_n \in \mathcal{S}_n^-(s_j), a \in \mathcal{T}_0, n \in \{j, \dots, N\}}. \end{aligned} \quad (56)$$

Consider the two following cases (enumerated as (iii) and (iv)) with respect to the strategy from which A deviates:

- (iii) If the strategy of A in Equation (56) matches the one obtained from Algorithm 2 on stage j , i.e., $x'^A(s_j) = x^{\circ A}(s_j)$ (note we are explicitly denoting A as the leader in stage j since $g_j < 0$)

$$V'_j(s_j, x_j^{\circ A}(s_j), x_j^{\circ B}(s_j, x_j^{\circ A}(s_j))) = V'_j(s_j, x_j^{\circ A}(s_j), x_j'^B(s_j, x_j^{\circ A}(s_j))),$$

and therefore Equation (50) holds. Then, for all other states, $s_n \in \mathcal{S}_n(s_j)$ for $n > j$, Equations (50) and (52) hold because of the hypothesis induction. Then, player A has no incentive to deviate in this case.

- (iv) If now the strategy of A in Equation (56) does not coincide with the one obtained by Algorithm 2 at stage j , i.e., $x'^A(s_j) \neq x^{\circ A}(s_j)$, then:

$$\begin{aligned} V'_j(s_j, x_j^{\circ A}(s_j), x_j^{\circ B}(s_j, x_j^{\circ A}(s_j))) &= \min_{a \in \mathcal{T}_0} V'_j(s_j, a, x_j^{\circ B}(s_j, a)) \\ &\leq V'_j(s_j, x_j'^A(s_j), x_j^{\circ B}(s_j, x_j'^A(s_j))), \end{aligned}$$

where the equality is because of Line 12 of Algorithm 2, and the inequality is because we are considering a particular action for A , $x_j'^A(s_j)$. Then, A has no incentive to deviate at stage j (at state s_j).

Since no player has an incentive to deviate, the solution obtained from Algorithm 2 is a feedback Stackelberg Equilibrium from a any state s_j with $j \in \{1, \dots, N\}$; in particular, for all $s_1 \in \mathcal{S}_1$, concluding the proof.

Appendix E: Proof of Proposition 1

We first show that for each stage $n \in \{1, \dots, N+1\}$, the payoff at each state s_n , $V_n''^*(s_n)$, is such that the probability of winning for player A is given by $1/2 + V_n''^*(s_n)/2$. By using induction in reverse order, note that this holds for stage $N+1$, i.e., in the border condition, where the race outcome is known by Equation (43). For the induction step, assume that for stage n , it holds that $1/2 + V_n''^*(s_n)/2$ is the winning probability for player A at state s_n . Let us consider any arbitrary state s_{n-1} in stage $n-1$. Note that by Equation (34), we have that:

$$\begin{aligned} \frac{1 - V_{n-1}''^*(s_{n-1})}{2} &= \frac{1 - \mathbb{E}_{W_{n-1}}[V_n''^*(s_n)]}{2} \\ &= \mathbb{E}_{W_{n-1}} \left[\frac{1 - V_n''^*(s_n)}{2} \right]. \end{aligned} \quad (57)$$

Using the induction hypothesis, the argument inside the expectation in Equation (57) corresponds to the probability of winning for player A . Then, the expectation of this will also have the same interpretation, i.e., showing the result for stage $n-1$. This concludes the induction. Then, applying the mixed equilibrium strategies of the first lap to the zero-sum payoff matrix U'' , we get that the winning probability of player A is given by:

$$(\pi^{A'*})^T \left(\frac{\mathbf{1}_{|\mathcal{T}| \times |\mathcal{T}|} - U''}{2} \right) \pi^{B'*} = \frac{1 - (\pi^{A'*})^T U'' \pi^{B'*}}{2},$$

where $\mathbf{1}_{|\mathcal{T}| \times |\mathcal{T}|} \in \mathbb{R}^{|\mathcal{T}| \times |\mathcal{T}|}$ is a square matrix with ones.

Appendix F: Instance size and solving time

Lap	Number of states	Solving time [s]
1	9	0
5	12,607,200	12
10	50,428,800	47
15	113,464,800	107
20	201,715,200	189
25	315,180,000	296
30	453,859,200	428
35	582,956,928	541
40	702,865,408	653
45	793,469,152	729
50	861,548,032	1,016
52	889,564,032	1,046
Total	20,478,351,241	5h 34' 23"

Table 5 Number of states and solving time per lap for the instance solved in Section 5.1

Appendix G: Alternative scenario with yellow flag

We show a scenario from the simulations mentioned in Section 5.1 in which a yellow flag benefits player *A*. Figure 7 shows a race scenario with a SC in laps 25 to 27. We observe that at the time of the SC, the best tire to choose (until the end of the race) is the Medium compound. Player *A* acts in accordance with this, whereas player *B* does not. During the SC, *B* changes to the Soft compound since he must choose a tire compound different from the Medium (because of the rule that a car must use at least two different tire compounds). Although player *B* is faster than *A* between laps 30 and 40, this changes after lap 40, since the Soft compound starts to degrade. As a result, *A* wins the race by 5 seconds.

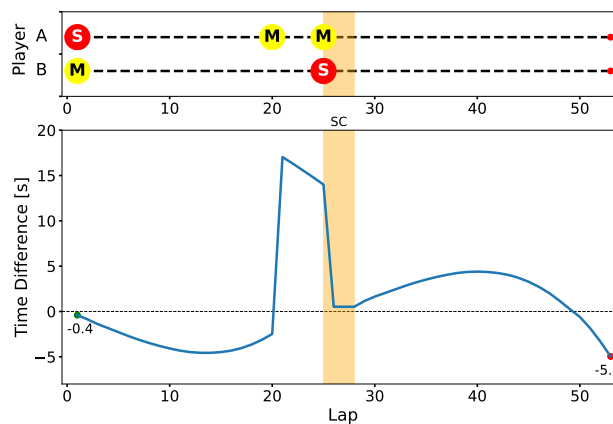


Figure 7 Race strategies of players (Top), partial race time difference (Bottom), scenario with SC in laps 25, 26 and 27.

Appendix H: Example of maximization winning odds

We show a race scenario exactly as the one shown in the right panel of Figure 4, when players maximize their winning odds, except that there is a VSC in lap 39. Figure 8 shows the players' strategies during the race in the top, and the partial time differences during the race in the bottom. It can be seen that in lap 39, when the VSC occurs, player *B* is ahead of *A*. Given that both players' tire compounds have more than 20 laps of wear, and there are 12 laps left until the end, both players decide to stop for Soft tire compounds. Then, *B* manages to keep the lead of the race from the VSC until the end of the race.

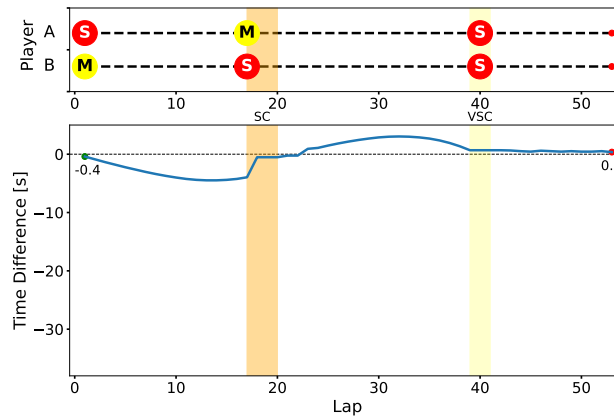


Figure 8 Race strategies of players (Top), partial race time difference (Bottom), scenario with SC in laps 17, 18 and 19; and VSC in laps 39 and 40.

Appendix I: Nash Equilibrium when Maximizing Probability of Winning

Table 6 shows the expected zero-sum payoff (U'') in the first lap for all combinations of initial tire compounds when the players maximize the probability of winning the race. The probabilities of the mixed strategy equilibrium of the first stage ($\pi^{A'*}$, $\pi^{B'*}$) are provided in the right column (bottom row) of Table 6 for player *A* (*B*).

		Player B				
		Tire	Soft	Medium	Hard	$\pi^{A' *}$
Player A	Soft		-0.14	-0.66	-0.64	0.40
	Medium		-0.48	-0.13	-0.65	0.60
	Hard		0.47	0.47	-0.13	0.00
		$\pi^{B' *}$	0.61	0.39	0.00	

Table 6 Expected players' payoff at the beginning of the race, U'' ; and mixed Nash Equilibrium, $\pi^{A'*}$, $\pi^{B'*}$